

BizRadar v4

Universal Sector Algorithm Specification

Batch 4 — Standalone Add-On

Sectors 31-33, 42, 48-49, 56 — Manufacturing, Wholesale Trade, Transportation & Warehousing, Administrative & Support / Waste Management. The B2B-heavy roster.

Read in conjunction with:

- Batch 1-2 PDF — Part 1 Architecture + Sectors 11/21/22/44-45/62/71/72/81 (114 pages cumulative)
- Batch 3 PDF — Sectors 23 Construction, 52 Finance, 53 Real Estate, 54 Professional Services (69 pages standalone)

This Batch 4 PDF contains 4 new sector chapters at the same depth + structure as Batches 1-3. The universal architecture, Master Router, and quality-gates content lives only in the Batch 1-2 PDF — this document assumes the reader has already read those.

BATCH 4 CONTEXT

Where Batch 4 fits

BizRadar v4 ships across 5 batches. Batches 1-2 delivered Part 1 (architecture, Master Router, universal scaffolding, quality gates) plus the Phase 0-1 sector roster — the v3 sweet spot (Sector 72 Hospitality) plus Sectors 11, 21, 22, 44-45, 62, 71, 81. Batch 3 covered the Phase 2 + Phase 3 professional + finance roster — Construction, Finance & Insurance, Real Estate, and Professional Services.

Batch 4 — this document — covers the Phase 4 B2B-heavy roster: the operational backbone of the US economy. Manufacturing (Sector 31-33) is the biggest sector by NAICS code count in the entire spec — 346 codes — and includes everything from a craft cheese maker selling DTC to a tier-2 auto parts supplier on a Boeing 737 program. Wholesale Trade (Sector 42) is the pure B2B middle layer that v3 doesn't even know exists. Transportation & Warehousing (48-49) is the invisible backbone every other sector depends on. Admin & Support / Waste Management (56) rounds out the recurring-revenue B2B roster — staffing, janitorial, security + pest, waste.

Batch 5 will deliver the remaining sectors (51 Information, 61 Education, 55 Holding Companies, 92 Public Administration) plus Parts 3-5 (NAICS-4 Failure-Rate Database, Implementation Roadmap, and Appendices).

What is in this Batch 4 PDF

Chapter	Coverage
Ch. 31-33 — Manufacturing	3 sub-clusters: food/bev DTC / industrial B2B / advanced regulated
Ch. 42 — Wholesale Trade	2 sub-clusters: durable goods / nondurable food + grocery
Ch. 48-49 — Transportation & Warehousing	3 sub-clusters: trucking / courier last-mile / warehousing 3PL
Ch. 56 — Admin & Support / Waste	4 sub-clusters: staffing / facility services / security + pest / waste

Reading order recommendation

Read 31-33.A food + beverage DTC first if you're building Phase 4 — it's the closest cousin to the v3 hospitality + retail logic from Batches 1-2 and the biggest small-business addressable in manufacturing. Then read 56.B facility services + 56.C pest control — high-multiple-EBITDA B2B-recurring-revenue businesses with the most M&A; activity in this batch. Then 48-49.A trucking + freight if you're covering owner-operator economics. Save 31-33.C advanced regulated manufacturing and 48-49.C warehousing 3PL for Phase 5 — they exercise long-cycle B2B-relationship + program-qualification patterns that don't appear elsewhere in the spec.

Quality bar — same as Batches 1-3

Every sub-cluster opportunity-playbook entry contains a NAMED real-world entity — real cert bodies (ISO 13485, AS9100, NADCAP, IATF 16949, NPMA, BSCAI, ISSA, NWRA, NCCO), real software platforms (Plex, Epicor, MachineMetrics, Werum PAS-X, Manhattan Associates, Locus Robotics, Onfleet, Bullhorn, PestPac, Soft-Pak), real federal programs (CMMC, DLA, GSA Schedule, Buy America, MEP Center), real industry associations (NAW, IFDA, IWLA, ATA, NPMA, ASA, BSCAI, NWRA, SWANA), and real benchmark sources. Engineering teams can look up every named entity to verify it's currently operating before encoding it into the `sector_profiles.js` file.

CHAPTER 31-33 — MANUFACTURING

Food + beverage producers, industrial B2B contract manufacturers, advanced + regulated manufacturers (medical device, aerospace, auto parts). Phase 4 priority — biggest sector + most operationally diverse.

One-line truth

A craft cheese maker selling DTC + farmers markets + Whole Foods has hospitality + retail logic. A tier-2 auto parts supplier sells to 6 named OEMs and the entire business runs on PPAP + APQP + IATF 16949 — concepts v3 has zero awareness of. A medical device manufacturer is regulated by FDA 510(k) + ISO 13485 and cannot do half the marketing v3 suggests for hospitality. Three radically different profiles inside one NAICS sector.

NAICS code count: 346 **Sub-clusters:** 3

manufacturing.food_beverage_dtc — Food + beverage producers (artisan, craft, farm-direct, DTC, regional brand)

Business model	B2C_online + B2B_wholesale hybrid
NAICS examples	311xxx (food mfg), 312xxx (beverage mfg)
Modern variants	Craft Cheese Maker, Specialty Coffee Roaster, Hot Sauce Brand, Craft Distillery, Specialty Bakery, Maple Syrup / Honey Producer
Competitor radius	DTC online: no physical radius (national). Local farmers market + tasting room: 30-60 mi. Regional grocery + specialty retail: state-wide + 4-state region. Distributor-driven: matches distributor coverage map.

Data sources

Source	Returns	Cost	Fallback
Google Places + Reviews (retail/tasting room)	Reviews + photos + hours	\$0.02	Yelp Fusion
Shopify / WooCommerce admin	DTC conversion + AOV + repeat-rate	Internal	Direct export
Faire + Mable + Tundra	Wholesale marketplace performance	Subscription	Direct
UNFI + KeHE + DPI Specialty Foods directories	Named distributor relationships	Direct	Trade show
Whole Foods Local Forager + named regional grocery buyers	Named retail buyer programs	Direct application	Buyer-direct

Source	Returns	Cost	Fallback
Specialty Food Association (SFA) + Good Food Foundation	Industry benchmarks + trade events	Member-only	Public summary
USDA + FDA + state ag dept	Compliance + inspection requirements	Free public	Direct
Farmers market manager + state ag tourism	Named market + farm-tour partnerships	Free / direct	Direct

Review categories tracked

- Product taste / quality
- Authenticity / origin story
- Packaging design + premium feel
- Shipping speed + condition (perishable)
- Pricing vs perceived value
- Sustainability / sourcing claims
- Allergen / dietary clarity (gluten-free, organic, etc.)
- Customer service / response

Opportunity playbook (16 categories)

Category	Research basis	Example with named entity
Specialty Food Association (SFA) + Fancy Food Show	Named trade events + buyer connections	Winter + Summer Fancy Food Show booth — \$4,800 booth + travel — captures 60-120 wholesale buyer leads + Whole Foods regional + Bristol Farms + Erewhon meetings
Whole Foods Local Forager program entry	Whole Foods Local Producer Loan + Local Forager submission	Whole Foods 5-store regional pilot — 800 SKUs/wk × \$4 wholesale × 5 stores = \$16k/wk = \$832k/yr potential at scale
UNFI / KeHE distribution agreement	Named natural-channel distributors + slotting fees	KeHE Naturally distribution — 800 store touchpoints — adds \$48k/mo at scale (after 18-mo ramp + slotting investment)
Faire + Mable wholesale marketplace listing	Named wholesale platforms + retailer profile	Faire Pro listing + retailer-of-the-month feature — captures 14-22 new wholesale accounts/mo at \$0 sales-rep cost
DTC Shopify + lifecycle marketing	Klaviyo + Postscript + named gift-box benchmarks	Shopify + Klaviyo welcome + abandoned cart + post-purchase + gift-box upsell — DTC revenue from \$2k/mo to \$48k/mo over 18 mo
Farmers market + festival circuit	Named regional markets + festival application timing	12-week farmers market circuit (Madison, Milwaukee, Chicago) — \$14k/wk gross — builds regional brand + retail-buyer interest

Category	Research basis	Example with named entity
Tasting room / on-site retail experience	Brewery + winery + creamery tasting-room economics	Tasting room build — captures 4-6× retail margin + drives email-list growth + tour fees
Awards + named-press recognition	Named awards (Good Food Awards, sofi, James Beard, Garden + Gun Made in the South)	Good Food Award + James Beard Foundation press — drives 18% wholesale-buyer interest + \$24k earned media value
Co-packing + private-label B2B line	Co-pack manufacturing economics + named clients	Co-pack contract with named retail brand — \$48k/mo recurring B2B revenue at 22% margin (vs. higher-margin DTC)
Subscription / club program	Named subscription platforms (Recharge, Bold, Stay AI)	\$45/mo wine + cheese club — 320 subscribers = \$173k/yr predictable revenue + 4× higher LTV
Regional grocery buyer program	Named regional chains (Trader Joe's, Sprouts, Wegmans, Publix, GreenWise, HEB Central Market, Heinen's)	Trader Joe's buyer pitch + 4-store pilot — adds \$14k/wk × 50 wk = \$700k/yr potential
Farm-to-table restaurant relationships	Named regional farm-to-table restaurants	14 named regional restaurants × \$1,200/mo each = \$16.8k/mo recurring + builds chef-network referral
Co-op + buying-group membership	Named natural co-ops (Frontier, NCG, INFRA)	NCG (National Co+op Grocers) approved vendor — captures 200+ member co-ops nationally
Sustainability + cert program (USDA Organic, Non-GMO Project, Fair Trade, B Corp, Kosher)	Named cert bodies + their pricing + market value	USDA Organic + Non-GMO Project Verified — \$4-12k/yr cert cost — captures 14% premium retail price + Whole Foods qualification
Influencer + UGC + recipe content engine	Named food bloggers + IG creator economics	Named regional food creators (Madison Eats, Milwaukee Foodie) — comp \$200 product → IG post → 80 site visits + 18 conversions
Co-packer scale-up vs in-house production	Co-pack economics + named co-packers	Move from in-house to named co-packer (e.g. SunOpta, ICONIC) — drops cost-per-unit 38% at scale + frees capital for marketing

Specificity rules (anti-hallucination)

- Every named distributor must be real (UNFI, KeHE, DPI, Sysco, US Foods, Whole Foods, Bristol Farms, Erewhon, Trader Joe's, Sprouts, Wegmans, Publix, Kroger, HEB, Heinen's)
- Every named cert body must be real (USDA Organic, Non-GMO Project, Fair Trade USA + Fair Trade International, B Corp, Kosher OU + OK + Star-K, Certified Humane, Animal Welfare Approved)
- Every named association must be real (Specialty Food Association, Good Food Foundation, NCG, INFRA, Frontier, BJCP for beer)

- Every named platform must be real (Shopify, WooCommerce, BigCommerce, Klaviyo, Recharge, Faire, Mable, Tundra)
- Every named regulatory framework must be real (FDA FSMA, USDA FSIS, state ag dept inspection, GFSI / SQF / BRC + cGMP)

Cost scale calibration

Food + beverage producer spend bands operate at small-business + wholesale-channel scale:

Band	Spend type
1 — Under \$500	Marketplace listing, partnership outreach, social content
2 — \$500 to \$5,000	Trade show booth, packaging refresh, cert application
3 — \$5,000 to \$50,000	Equipment add, broker / sales-rep hire, slotting fee
4 — \$50,000 to \$500,000	Co-packer setup, full DTC launch, distribution slotting
5 — Over \$500,000	New facility, brand acquisition, national launch

Persona schema

B2B (wholesale buyer) + B2C (DTC consumer) hybrid

- Buyer type (DTC consumer / specialty retailer / regional grocer / national chain / restaurant / co-pack)
- Decision driver (price, quality, story, margin, exclusivity)
- Acquisition channel (trade show, broker, marketplace, DTC ad, referral)
- Reorder cadence + volume
- Promotion / slotting + co-op marketing expectation
- LTV (monthly volume × margin × retention years × cross-sell SKUs)

Cycle / calendar model

Food + beverage runs on harvest + holiday + wholesale-cycle + festival cycles:

Window	Trigger	Action
Q4 holiday (Sep-Dec)	40-60% of annual DTC revenue (gift-driven)	Holiday SKUs + gift packaging + maximum DTC spend
Spring (Mar-May)	New-product launches + farmers market opening	Trade show + new-SKU launch + market debut
Summer (Jun-Aug)	Peak farmers market + tourism + festivals	Festival circuit + tasting room + tourism push
Trade show season (Jan + Jun)	Winter + Summer Fancy Food Show + Expo East/West	Booth + buyer meetings + new-SKU debuts
Production cycle (sector-specific)	Harvest + maple-tap + seasonal	Production scheduling + inventory build
Wholesale buyer cycles	Buyer reviews 1-2× per year per chain	Slotting + buyer-relationship cadence

Window	Trigger	Action
Annual: cert + inspection renewal	USDA + FDA + state + cert	Compliance + audit prep

Issue library

- Slotting fee + new-product introduction cost — universal — frame as DTC + Faire + small-account first
- Distributor margin pressure (UNFI, KeHE chargebacks) — universal — frame as direct + brand-side discipline
- Cold-chain + shipping-loss in DTC — universal — frame as named carrier + insulated packaging
- Capacity constraints in scale-up — universal — frame as co-packer transition
- Price increases pass-through to retailers — universal — frame as transparent comm + cost backup
- Cert program complexity — universal — frame as multi-cert audit consolidation
- Co-packer reliability + quality — universal — frame as audit + named alternatives
- Regional vs national strategy decision — universal — frame as deliberate channel sequence

Report sections

Section	Contents
Producer snapshot	Categories, production capacity, SKU count, channel mix, named comp set
Top 5 channel + scale moves	Drawn from 16-category playbook
Wholesale channel strategy	Distributor + retail + Faire + co-pack opportunities
DTC + lifecycle marketing plan	Named platforms + flow architecture
Trade show + buyer-meeting pipeline	Named events + target buyers
Cert + sustainability roadmap	Named programs + audit prep
Customer + buyer personas (4 + lost account)	DTC / specialty / regional / chain
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + channel mix %
- Wholesale account count + reorder rate
- DTC AOV + LTV + repeat rate
- Production capacity utilization %
- Slotting + co-op marketing % of wholesale revenue
- Cert / audit pass rate (target 100%)
- Sell-through rate at key retailers

manufacturing.industrial_b2b — Industrial B2B contract manufacturers (machine shops, fabrication, plastics, metal stamping, electronics assembly)

Business model	B2B_relationship
NAICS examples	332xxx (fabricated metal), 333xxx (machinery), 334xxx (electronics — non-medical), 335xxx (electrical equipment)
Competitor radius	Local / regional: 200-400 mi for shipping economics. Specialty / niche capabilities: national. Federal contract: national. Mexican / Asian competition for commodity work: global. Domestic re-shoring trend favors regional.

Data sources

Source	Returns	Cost	Fallback
ThomasNet + GlobalSpec	B2B industrial directory + RFQ sourcing	Subscription	Industry Net
Xometry + Hubs (Protolabs) + Fictiv marketplace	On-demand mfg marketplaces + lead flow	Free / sub	Direct
Manufacturers + named industry directories (NAM, NTMA, PMA, AWS)	Named peer competitors	Member-only	Public
LinkedIn Sales Navigator + Apollo	B2B prospect intelligence (engineers + buyers)	Subscription	LinkedIn
ISO + AS + IATF cert databases	Quality cert tracking + verification	Free / member	Direct
Federal SAM.gov + DLA + GSA	Federal contract opportunities	Free	Direct
Named MES + ERP platforms (Plex, Epicor, IQMS, NetSuite, Infor)	Internal benchmarks + production data	Subscription	Internal
MFG Day + IndustryWeek + Manufacturing.gov	Industry trends + workforce + benchmarks	Free	Public

Review categories tracked

- On-time delivery rate
- First-pass-yield / quality (PPM defect)
- RFQ response speed
- Engineering + DFM (design for manufacturability) support
- Pricing competitiveness
- Communication during production
- Cert + compliance reliability

- Capacity flexibility

Opportunity playbook (15 categories)

Category	Research basis	Example with named entity
ISO 9001 + AS9100 + IATF 16949 quality certification	Named cert bodies (DNV, BSI, NSF-ISR, Smithers, NQA, Lloyd's Register) + audit cycle	AS9100D for aerospace adjacent + ISO 9001:2015 — captures aerospace + tier-2 OEM work at 22% premium \$/hr vs commodity machine shops
On-demand marketplace (Xometry, Protolabs Hubs, Fictiv) listing	Named platforms + capacity + listing economics	Xometry Partner Network — captures 14-28 new RFQs/wk on idle capacity at 12% commission to Xometry
CAD / CAM / MES upgrade	Named platforms (Mastercam, Fusion 360, SolidWorks CAM, GibbsCAM, Plex MES, Epicor)	Mastercam + Plex MES integration — 18% productivity lift + real-time WIP tracking + drops planned-vs-actual variance
Industry 4.0 + IIoT + machine monitoring	Named platforms (MachineMetrics, FORCAM, Siemens MindSphere, GE APM)	MachineMetrics on 12 CNC machines — equipment utilization from 38% to 62% — captures \$480k/yr in lost-capacity recovery
Niche industry vertical specialization	Named verticals (aerospace, medical, defense, food + bev, oil + gas, EV)	EV battery component specialty — captures growing market + premium \$/hr — only 12 regional competitors
Federal / DoD / DLA prime + sub work	Named programs (DLA Land + Maritime, DoD Mentor-Protégé, Defense Contract Management Agency, ITAR)	ITAR registration + DLA prime registration — qualifies for \$14M annually in defense work in named region
CMMC + cybersecurity compliance (defense supply chain)	CMMC 2.0 + NIST 800-171 + named C3PAO assessors	CMMC Level 2 cert with named C3PAO — qualifies for prime + sub roles on \$4M+ DoD work otherwise inaccessible
Engineering + DFM as paid offering	Front-end design services + their economics	In-house engineering team + paid DFM service — captures \$80k/yr in design fees + locks production work for 5+ yrs per program
Named-OEM tier-2 supplier qualification	Named OEM supplier programs (GM, Ford, Stellantis, Toyota, Honda, John Deere, Caterpillar, Boeing)	Tier-2 supplier qualification with named OEM — captures \$4-12M/yr program-based revenue at 8-15% margin

Category	Research basis	Example with named entity
Lean + Six Sigma + Kaizen culture	Named training programs (TWI, ASQ, Lean Enterprise Institute, Kaizen Institute)	Lean transformation + 5S + value stream mapping — drops cost 18% + improves on-time delivery from 84% to 96%
Tariff + reshoring positioning + Buy America	Named federal programs (Buy America Act, BABA, Reshoring Initiative, MEP Center)	Buy America Act compliance + MEP Center engagement — captures domestic-supply-chain RFQs + federal-aid work premium pricing
Apprenticeship + workforce development	DOL Registered Apprenticeship + named technical colleges + NIMS	NIMS-Registered Apprenticeship with named technical college — 6 apprentices/yr at federal grant subsidy + solves talent shortage
Multi-shift + capacity expansion	Shift premium economics + capital investment	Add second shift on bottleneck machines — 60% capacity lift at 15% incremental labor cost — drives growth without capex
Named-distributor + sales-rep network	Named manufacturer reps + distributor partnerships	3 named manufacturer rep firms in 4-state territory — captures 28% of new RFQs at 8-12% commission
Capability-based marketing (precision tolerances, materials, processes)	Named capability marketing + ThomasNet + GlobalSpec	"Tolerance to ± 0.0001 " + 5-axis CNC + Inconel + titanium" specialty — captures 14% premium \$/hr from aerospace + medical work

Specificity rules (anti-hallucination)

- Every named cert must be real (ISO 9001, ISO 14001, ISO 13485, AS9100, AS9120, IATF 16949, NADCAP, ITAR, CMMC, NIST 800-171, RoHS, REACH)
- Every named platform must be real (Plex, Epicor Kinetic, Infor LN, IQMS, NetSuite Manufacturing, JobBOSS², Global Shop, ECi M1, MachineMetrics, FORCAM)
- Every named CAD/CAM must be real (SolidWorks, AutoCAD Inventor, Fusion 360, Mastercam, GibbsCAM, NX, CATIA, Siemens NX)
- Every named association must be real (NAM, NTMA, PMA, AWS, ASM International, SME, SAE)
- Every named OEM must be a verifiable currently-active manufacturer in that vertical
- Every named federal program must be real (Buy America, BABA, DLA programs, MEP Center, MAPP, Manufacturing USA institutes)

Cost scale calibration

Industrial B2B mfg spend bands operate at capital-equipment + technology + cert scale:

Band	Spend type
1 — Under \$5,000	Marketplace listing, association membership, training course
2 — \$5,000 to \$50,000	Cert application, software stack add, marketing

Band	Spend type
3 — \$50,000 to \$500,000	CAD/CAM upgrade, MES platform, single-machine add
4 — \$500,000 to \$5M	Major equipment line, second shift hire, capability-vertical buildout
5 — Over \$5M	New facility, M&A; acquisition, full re-equipment

Persona schema

B2B_buyer (engineer + buyer + program manager committee)

- Buyer role (program mgr, supply chain, design engineer, quality engineer)
- Sourcing process (RFQ, sole-source, prequalified-list, marketplace)
- Decision driver (price, on-time, quality / PPM, capability, cert)
- Industry vertical match
- Volume tier (prototype / low-volume / mid-volume / high-volume)
- Annual program value + 3-7 yr contract LTV

Cycle / calendar model

Industrial mfg runs on automotive model-year + aerospace program + capital-budget cycles:

Window	Trigger	Action
Automotive model-year cycle	New model program cycle 4-7 yrs	Tier-2 supplier qualification + APQP launch + PPAP
Aerospace program cycle	Long programs 10-30 yrs (Boeing 737, F-35)	Long-cycle program qualification + AS9100 + sustainment
Federal FY-end (Sep)	Federal spending deadline	DLA + DoD bid pipeline ramp
Capital budget season (Aug-Nov)	Customer capital plan finalized	Engagement on next-year programs + design-in opportunity
Annual: cert audit cycle	ISO + AS + IATF surveillance + recert	Pre-audit walkthrough + management review
Quarterly: customer QBR + scorecard	Customer-managed scorecard	On-time + PPM + responsiveness review
Trade show season (Jan-Mar + Sep-Oct)	Named industry shows (IMTS, EASTEC, FABTECH)	Booth + customer meetings + new-business pipeline

Issue library

- On-time delivery / customer scorecard pressure — universal — frame as MES + capacity discipline
- PPM defect rate / quality complaints — universal — frame as Six Sigma + 8D + cert systems
- RFQ response speed — universal — frame as estimating discipline + named platforms
- Engineering / DFM resource constraints — universal — frame as paid engineering line
- Capacity vs commodity-pricing pressure — universal — frame as niche + premium positioning
- Cert audit failure risk — universal — frame as continuous internal audit + management review

- Talent / craft labor shortage — universal — frame as apprenticeship + automation
- Tariff + supply-chain volatility — universal — frame as Buy America + reshoring positioning

Report sections

Section	Contents
Mfg snapshot	Capabilities, equipment list, certs, vertical mix, annual revenue, named comp set
Top 5 capability + program moves	Drawn from 15-category playbook
Quality + cert roadmap	Named cert programs + audit prep
Industry vertical specialization plan	Named OEM + tier-2 supplier qualification pathway
Tech stack + Industry 4.0 roadmap	Named MES + IIoT + monitoring priorities
Federal + DoD + reshoring opportunity	Named programs + qualification roadmap
Customer scorecard + retention plan	Named OEM customers + QBR cadence
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per FTE
- On-time delivery % (target > 96%)
- First-pass yield + PPM defect rate
- Equipment / shop utilization %
- Customer scorecard rating from named OEMs
- New-program win rate
- Profit margin (target > 10% on machine work, > 18% on assembly + value-add)

manufacturing.advanced_regulated — Advanced + regulated manufacturing (medical device, pharmaceutical, aerospace prime, semiconductor, advanced auto)

Business model	B2B_relationship + regulated
NAICS examples	325xxx (chemicals + pharma), 334510 + 339113 (medical instruments + devices), 336xxx (aerospace + auto + transportation equipment), 334412/334413/334419 (semiconductor)
Competitor radius	National + global. Regulated mfg competes globally on capability + cert + price. Domestic preference in defense + medical device. Specialty / niche national. Most highly regulated mfg has 4-12 global competitors per program area.

Data sources

Source	Returns	Cost	Fallback
FDA 510(k) + PMA + DMF database	Named regulated competitors + product clearances	Free public	Direct
Named regulator databases (FDA, EPA, OSHA, FAA, NHTSA, NRC)	Compliance + recall + approval status	Free public	Direct
ISO + AS + IATF cert databases	Quality cert tracking	Free / member	Direct
SAM.gov + Federal procurement	DoD + DOT + NASA + DHS opportunities	Free	Direct
EU MDR + CE marking + named EU notified bodies	EU + global market access	Free / sub	Direct
Industry trade publications (MedTech Insight, Aviation Week, EE Times, SemiWiki)	Industry trends + competitive intel	Subscription	Public
LinkedIn + Apollo + ZoomInfo (named buyer + program mgr)	B2B prospect + customer intel	Subscription	Direct
Named industry associations (AdvaMed, MedAccred, AIA, SEMI, RIA)	Industry benchmarks + named peers	Member-only	Public summary

Review categories tracked

- Regulatory compliance + audit history
- Quality + PPM defect (sigma-level)
- On-time delivery + program-milestone reliability
- Engineering / R&D; capability
- Documentation + traceability
- Validation + verification execution (IQ/OQ/PQ)
- Cost competitiveness vs offshore
- Long-term program reliability

Opportunity playbook (15 categories)

Category	Research basis	Example with named entity
ISO 13485 + FDA QSR + EU MDR compliance maturation	Named registrars (BSI, TÜV SÜD, Dekra, NSAI, SGS)	ISO 13485 + EU MDR notified-body audit pass — captures EU market access + regulator-driven customer preference at 28% premium \$/unit

Category	Research basis	Example with named entity
AS9100 + AS9120 + NADCAP for aerospace	NADCAP commodity-specific accreditation	NADCAP for chemical processing + heat treat + welding — captures aerospace tier-1 + tier-2 work otherwise inaccessible
IATF 16949 + automotive customer-specific requirements (CSRs)	Named OEM CSRs (FCA, GM, Ford, Toyota, Honda)	IATF 16949 + Ford Q1 + GM BIQS + Toyota CSRs — qualifies for tier-1 work on \$40M annual programs
CMMC + NIST 800-171 + DFARS for defense	Defense supply chain compliance + named C3PAO	CMMC Level 2 + named C3PAO assessment — qualifies for \$40M+ DoD prime + sub work otherwise blocked
Validation + V&V; process discipline (IQ/OQ/PQ)	Named V&V; tools + protocols + training (ASQ-CSSBB, ASQ-CQE)	Validation Master Plan + IQ/OQ/PQ protocol library — drops audit findings + drops time-to-market by 30%
Manufacturing Execution System (MES) + electronic batch records	Named platforms (Honeywell Tridium, Werum PAS-X, MasterControl, Apprentice)	Werum PAS-X for pharma — eliminates paper batch records + drops batch-release time 60% + supports global regulatory audits
Named-OEM tier-1 supplier qualification	Named OEM supplier qualification programs (Boeing, Lockheed Martin, Raytheon, Northrop, GE Aviation, Pratt & Whitney, Medtronic, J&J, BD, Stryker)	Boeing tier-1 qualification + Boeing Quality Management System (BQMS) — captures \$20-100M annual program revenue at 12-22% margin
Cleanroom + ISO 14644 environmental control	Named cleanroom suppliers + classification (ISO 14644-1 Class 7 + 8 + 9 + Class 5 / 100)	ISO 14644-1 Class 7 cleanroom build — captures medical device + sterile-pharma + semiconductor work — premium \$/sqft mfg
FDA 510(k) submission expertise + RA/QA infrastructure	Named RA/QA consultancies + 510(k) success rates	In-house RA/QA team + named consultancy partner — drives 510(k) clearance in 9-12 mo vs industry avg 18-24 mo — captures earlier revenue
Predicate device + 510(k) pathway expansion	FDA 510(k) database + predicate device research	Multiple 510(k) submissions per yr leveraging predicate strategy — captures 4-6 product launches/yr at \$4-12M revenue each
CRO / contract development partnerships	Named CROs + CDMOs + named partners	Partnership with named CRO (Charles River, Eurofins, NAMS) — captures development-phase work + locks commercial mfg

Category	Research basis	Example with named entity
Pharma cGMP + FDA inspection readiness	cGMP + FDA inspection criteria + named consultants	cGMP + FDA pre-approval inspection readiness — captures CDMO partnership work otherwise blocked by inspection risk
Recall + post-market surveillance + complaint handling	FDA MAUDE + recall management	In-house complaint handling + post-market surveillance — drops recall risk + improves regulator relationship + drives customer retention
IP + patent portfolio + trade-secret protection	Named IP firms + patent prosecution	Strategic patent portfolio (12 patents per product family) — drives 28% premium pricing + freedom-to-operate + acquisition value
M&A; + strategic acquisition / divestiture	Named investment banks + valuation multiples (medical device 4-8× revenue, aerospace 8-12× EBITDA, pharma 10-15× EBITDA)	Strategic acquisition of complementary product line — captures product synergy + customer cross-sell + defensive moat

Specificity rules (anti-hallucination)

- Every named cert must be real (ISO 13485, ISO 14971, IEC 60601, IEC 62304, ISO 14644, FDA QSR 21 CFR 820, EU MDR 2017/745, IATF 16949, AS9100, NADCAP, cGMP 21 CFR 210/211)
- Every named regulator must be real (FDA, EPA, OSHA, FAA, NHTSA, NRC, EMA, MHRA, PMDA, NMPA, MDSAP)
- Every named OEM must be a verifiable currently-active prime in that vertical
- Every named platform must be real (MasterControl, Veeva Vault QMS, Werum PAS-X, ETQ Reliance, Sparta Trackwise, Apprentice, Honeywell Tridium)
- Every named association must be real (AdvaMed, MedAccred, AIA, SEMI, RIA, MedTech Innovator, BIO, PhRMA)
- Every named CRO/CDMO must be real (Charles River, Eurofins, NAMSA, Lonza, Catalent, Thermo Fisher, WuXi, IQVIA)

Cost scale calibration

Advanced regulated mfg spend bands operate at major-capital + multi-year-program scale:

Band	Spend type
1 — Under \$50,000	Cert application, association membership, RA consultant
2 — \$50,000 to \$500,000	V&V; infrastructure, MES module, software stack
3 — \$500,000 to \$5M	Cleanroom retrofit, equipment validation, regulatory submission
4 — \$5M to \$50M	Major facility expansion, FDA / CE submission, M&A;
5 — Over \$50M	New facility, major M&A;, plant build, multi-program expansion

Persona schema

B2B_buyer (regulated procurement + program mgmt + RA/QA committee)

- Buyer role (program mgr, RA/QA, supply chain, regulatory affairs)
- Sourcing process (long-cycle qualification, audit-driven, sole-source common)
- Decision driver (regulatory compliance, quality, on-time, IP, partnership)
- Program type + lifecycle stage (development / pre-launch / commercial / sustaining)
- Annual program value + 5-15 yr contract LTV
- Customer regulatory risk tolerance

Cycle / calendar model

Advanced regulated mfg runs on multi-year program + regulatory + audit cycles:

Window	Trigger	Action
Multi-year program cycles (3-15 yrs)	Long-cycle programs	Long-cycle BD + customer relationship + regulatory milestones
Annual: regulatory audit cycle (FDA, EU, IATF, AS)	Multiple registrar audits	Pre-audit prep + management review + CAPA closure
Quarterly: customer QBR + scorecard	Customer scorecard discipline	On-time + PPM + program milestone review
Submission cycles (510(k), PMA, MDR, type certification)	Regulatory clearance milestones	Submission discipline + RA/QA capacity
Federal FY-end (Sep)	DoD + NASA + DOE spending	Federal program bid pipeline
Trade show season	Named events (MD&M, Pharma Expo, Paris Air Show, Semicon)	Booth + customer meetings + new-program pipeline
Industry program lifecycle (model year, generational shift)	Multi-year industry shifts	Strategic positioning for next-gen programs

Issue library

- Regulatory audit findings + warning letters — universal — frame as continuous compliance + CAPA discipline
- PPM defect / scrap on validated processes — universal — frame as Six Sigma + V&V; + change control
- Long product-launch cycle vs market opportunity — universal — frame as predicate strategy + RA expertise
- Customer-specific requirements complexity — universal — frame as CSR system + customer-portal discipline
- Cert + audit cost burden — universal — frame as multi-cert audit consolidation + named registrars
- Reshoring + Buy America vs offshore cost competitiveness — universal — frame as automation + premium positioning
- IP + trade-secret protection vs talent mobility — universal — frame as IP discipline + employee agreements

- Cleanroom + facility validation maintenance — universal — frame as preventive maintenance + monitoring

Report sections

Section	Contents
Mfg snapshot	Capabilities, products, certs, vertical mix, annual revenue, named comp set
Top 5 strategic moves	Drawn from 15-category playbook
Regulatory + cert roadmap	Named cert + submission programs + audit prep
Industry vertical specialization plan	Named OEM + tier-1 qualification pathway
V&V; + quality + MES capability roadmap	Named platforms + investment priorities
Federal + defense + global market opportunity	Named programs + qualification roadmap
IP + M&A; + strategic positioning plan	Named IP firm + acquisition + divestiture pipeline
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per FTE
- On-time delivery % (target > 98%)
- PPM / sigma-level + audit findings count
- Customer scorecard from named OEMs
- Time-to-market on new program launches
- Regulatory submission success rate
- IP + patent count + freedom-to-operate position
- Profit margin (target > 18% on regulated work)

Engineering guidance

Sector 31-33 is the biggest by code count (346) and most operationally diverse. Build manufacturing.food_beverage_dtc first (Phase 4, biggest small-business opportunity, B2C + B2B hybrid most familiar). Industrial B2B (31-33.B) is Phase 4 (B2B-relationship + tier-2 supplier qualification). Advanced regulated (31-33.C) is Phase 5 specialty — long-cycle programs + global regulators + 4-12 competitors per program. Total addressable count of small mfg in US: ~250k businesses, vast majority in food/bev DTC + industrial B2B sub-clusters.

CHAPTER 42 — WHOLESALE TRADE

Industrial + machinery distributors, electronics distributors, food + grocery wholesalers, beverage distributors, building materials. Pure B2B middle layer between manufacturers and retailers.

One-line truth

A wholesaler doesn't have customers walking through the door. Their entire business is named-account books — 200-2,000 named retail / commercial customers — and the moats are line-card breadth + delivery reliability + credit terms. v3 doesn't even know wholesalers exist.

NAICS code count: 71 **Sub-clusters:** 2

wholesale.durable_goods — Durable goods distributors (industrial, electronics, building materials, machinery, automotive parts)

Business model	B2B_relationship
NAICS examples	423xxx (durable goods distribution — full sub-sector)
Competitor radius	Local: 100-300 mi for delivery economics. Regional: 4-state region. National distributors (Grainger, Fastenal, MSC) compete in every market. Specialty / niche distributors regional or national. Federal contract: national.

Data sources

Source	Returns	Cost	Fallback
Named ERP / DMS systems (SAP, Oracle, Microsoft Dynamics, Epicor Eclipse, Infor SXe, NetSuite)	Internal customer + product data	Subscription	Internal
ThomasNet + GlobalSpec (industrial)	B2B sourcing intelligence	Subscription	Industry Net
NAW (National Association of Wholesaler-Distributors)	Industry benchmarks + named peers	Member-only	Public summary
Industry-vertical associations (NEMA, NECA, ARW, IFDA)	Vertical-specific benchmarks + intel	Member-only	Direct
Modern Distribution Management (MDM)	Industry trends + benchmarks + Top 100 lists	Subscription	Public
LinkedIn + Apollo + ZoomInfo	Named buyer + customer intel	Subscription	LinkedIn
SAM.gov + GSA + DLA	Federal opportunities + specs	Free	Direct

Source	Returns	Cost	Fallback
Named manufacturer rep firms + line-card listings	Manufacturer line-card relationships	Direct	Trade show

Review categories tracked

- Line-card breadth + named brands carried
- Delivery reliability + same-day / next-day fill rate
- Pricing competitiveness + program flexibility
- Counter / branch / outside sales relationships
- Credit terms + AR flexibility
- Tech + e-commerce + EDI capability
- Returns + warranty handling
- Technical support + DFM advice

Opportunity playbook (15 categories)

Category	Research basis	Example with named entity
E-commerce + customer self-service portal	Named platforms (Unilog, Magento Commerce + Adobe, BigCommerce B2B, Sana Commerce)	Unilog Commerce + product data syndication — captures 28% of orders online + frees branch staff for value-add selling
Punchout + EDI + customer integration	Named B2B integration platforms (Cleo, SPS Commerce, TrueCommerce, OpenText)	EDI 850/855/856/810 with named top-20 customers — drops processing cost 80% per order + locks customer integration
Inventory + ERP + WMS modernization	Named platforms (Epicor Eclipse, Infor SXe, NetSuite, SAP S/4HANA, Oracle NetSuite, Acumatica)	Epicor Eclipse + RF scanner WMS — drops inventory inaccuracy from 8% to <1% + enables same-day fill rate target
Line-card + brand expansion	Named manufacturers + competitive analysis	Add 4 named brand line-cards (e.g. for industrial: Hilti, Milwaukee, DeWalt, Hubbell) — captures cross-sell + competitive position
Private label / store-brand line	Named contract manufacturers + private label economics	Private label tier — 60% gross margin vs 25% on branded — drives 18% of total profit on 8% of revenue
Outside sales territory + comp redesign	Named CRM platforms + territory mapping (Salesforce, MapAnything, Dynamics 365 Field)	Salesforce + named CRM + territory + activity tracking — drives 22% revenue lift from existing rep base via discipline

Category	Research basis	Example with named entity
Inside sales + telesales B2B development	Named platforms (Outreach, SalesLoft, Apollo, Gong)	Inside sales team + Outreach cadence — adds \$480k/yr revenue from 200 named house accounts at \$80 CAC
Vendor managed inventory (VMI) program	VMI economics + named programs	VMI program with named top-10 customer accounts — locks 5-yr contract + drives stock-fill premium pricing
Manufacturer rep agreement + exclusive territory	Named manufacturer rep programs	Exclusive territory rights from named manufacturer — captures premium % + competitive moat in territory
Branch + counter + showroom optimization	Branch profitability + customer-pickup economics	Branch consolidation + counter-pro program — drops OpEx 18% + concentrates volume + improves profitability
Federal + DLA + GSA Schedule	GSA Schedule + DLA Land + Maritime + DoD MAS programs	GSA Schedule listing + DLA prime contract — captures \$4M/yr predictable federal volume at 14% margin
Credit terms + AR optimization	Named credit insurance partners (Allianz Trade, Atradius, Coface, Euler Hermes)	Credit insurance + credit-management discipline — drops bad-debt write-offs + enables aggressive new-account terms
Technical service + value-add (kitting, assembly, custom)	Named value-add services + margin profile	Kitting + light assembly value-add — captures 38-50% margin (vs 22% on commodity distribution) + drives customer loyalty
Acquisition + roll-up strategy	Named M&A; platforms + valuation multiples (5-8× EBITDA typical)	Acquire smaller competitor at 6× EBITDA — adds revenue + line-card consolidation + customer cross-sell
Industry vertical specialization	Named industry verticals + named buyer programs	Healthcare-vertical specialty distributor — captures 22% premium \$/SKU + higher reorder rate + lower competition

Specificity rules (anti-hallucination)

- Every named ERP / DMS must be real (Epicor Eclipse + Prophet 21, Infor SXe + Distribution SX.e, NetSuite, SAP, Oracle, Microsoft Dynamics, Acumatica, Sage X3)
- Every named e-commerce platform must be real (Unilog, Magento + Adobe Commerce, BigCommerce B2B Edition, Sana Commerce, Insite Commerce, Shopify Plus B2B)
- Every named manufacturer rep firm must be real (named industry-specific firms vary — many regionally-named multi-line rep groups)
- Every named association must be real (NAW, MDM, plus vertical-specific: NEMA, NECA, NAED, ARW, IFDA, NAPA, ASA, AHF, AHRI)
- Every named credit insurer must be real (Allianz Trade, Atradius, Coface, Euler Hermes [now Allianz Trade], CNA, Trade Risk)

Cost scale calibration

Durable goods distribution spend bands operate at inventory + technology + branch scale:

Band	Spend type
1 — Under \$5,000	Association membership, training, marketing template
2 — \$5,000 to \$50,000	CRM + ERP module add, marketing campaign, EDI partner setup
3 — \$50,000 to \$500,000	E-commerce + ERP upgrade, branch refresh, sales team expansion
4 — \$500,000 to \$5M	New branch, major ERP migration, M&A; acquisition (small)
5 — Over \$5M	Major M&A;,, new region expansion, full ERP transformation

Persona schema

B2B_buyer (procurement + counter-pro + maintenance + commercial buyer)

- Buyer role (purchasing manager, contractor counter-pro, MRO buyer, project manager)
- Account size (small contractor / medium commercial / large industrial / national chain)
- Decision driver (price, line-card breadth, fill rate, credit terms, service)
- Acquisition channel (outside sales, walk-in counter, e-commerce, EDI, marketplace)
- Reorder cadence + volume + product mix
- LTV (annual revenue × margin × retention years × cross-sell)

Cycle / calendar model

Distribution runs on construction season + manufacturing cycle + federal FY cycles:

Window	Trigger	Action
Construction season (Apr-Oct in north)	Peak demand for building materials, MRO, electrical, plumbing	Capacity max + delivery discipline + counter-pro coverage
Federal FY-end (Sep)	Federal spending deadline	Federal account ramp + GSA + DLA orders
Annual: customer QBR + line-review	Customer-managed line reviews	QBR discipline + scorecard performance
Quarterly: customer pricing + program review	Customer pricing + rebate programs	Pricing discipline + program audit
Trade show season (named industry shows)	Industry events + new-product launches	Booth + customer meetings + manufacturer rep relationships
Inventory cycle (rolling)	Inventory turn + cycle counts	Inventory discipline + slow-moving SKU mgmt
Annual: ERP + tech roadmap	Tech investment cycle	Tech roadmap discipline

Issue library

- Inventory accuracy + cycle count discipline — universal — frame as RF scanner + WMS
- Margin compression from Amazon Business + national distributors — universal — frame as service moat + value-add
- Slow ERP / DMS modernization — universal — frame as named-platform roadmap
- Outside sales productivity vs cost — universal — frame as CRM + territory discipline
- Bad-debt write-offs from contractor failures — universal — frame as credit insurance
- Drop-ship + JIT pressure from customers — universal — frame as VMI + named integration
- Manufacturer rep relationship dynamics — universal — frame as line-card discipline + program performance
- Branch profitability + footprint optimization — universal — frame as data-driven consolidation

Report sections

Section	Contents
Distributor snapshot	Branches, employees, line-card, vertical mix, annual revenue, named comp set
Top 5 capability + scale moves	Drawn from 15-category playbook
E-commerce + EDI + tech roadmap	Named platforms + investment priorities
Line-card expansion + manufacturer rep plan	Named brand prospects + rep relationships
Sales team + territory optimization	Outside + inside + counter coverage
Customer + named-account pipeline	Named target accounts + QBR cadence
Vertical specialization + value-add plan	Named verticals + value-add services
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per branch
- Inventory turns (target > 6× / yr)
- Gross margin % (durable typical 22-32%)
- Customer fill rate (target > 95%)
- On-time delivery %
- New-account growth + customer retention
- E-commerce / EDI % of revenue
- Outside sales productivity (\$ revenue per rep)

wholesale.nondurable_food_grocery — Nondurable goods distributors (food + grocery wholesalers, beverage distributors, paper + packaging, chemicals + cleaning)

Business model	B2B_relationship
NAICS examples	424xxx (nondurable goods distribution — full sub-sector)
Modern variants	Specialty Food Distributor, Restaurant + Foodservice, Convenience Store Distributor, Janitorial Supply
Competitor radius	Foodservice + restaurant: 200-300 mi for daily delivery economics. Specialty / niche food: regional 4-state. Beverage distribution: state-defined (alcohol franchise law). Convenience: regional. National giants (Sysco, US Foods, PFG) operate national + regional split.

Data sources

Source	Returns	Cost	Fallback
Sysco + US Foods + Performance Food Group benchmarks	Industry benchmarks (named giants)	Public	Trade press
IFDA + IFMA + UNFI + KeHE + DPI	Named food distribution networks + benchmarks	Member-only	Public
Beverage Industry magazine + Beverage Daily	Beverage distribution intel	Subscription	Public
Named ERP / DMS (NetSuite, SAP, Encompass, Aldata, RELEX, Blue Yonder)	Internal benchmarks	Subscription	Internal
Convenience Store News + NACS	C-store distribution intel	Subscription / member	Public
Named foodservice GPOs (PFG Vistar, Premier, Vizient, FoodBuy)	Named GPO contracts + pricing	Direct	Industry
SAM.gov + DLA Troop Support	Federal food + grocery contracts	Free	Direct
Local restaurant + foodservice association	Named restaurant + chain prospects	Free / member	Direct

Review categories tracked

- Delivery reliability + on-time-in-full
- Order accuracy + fill rate
- Product quality (food-safety + freshness)

- Pricing + program rebates
- Sales rep + relationship
- Credit terms + AR flexibility
- Tech + ordering platform
- Returns + credit handling

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Online ordering + customer portal	Named platforms (eFoods Direct, Cuttwood, BlueCart, Choco, Notch, Foodbam)	BlueCart + Choco online ordering platform — captures 38% of orders online + drops customer-service phone time 60%
Foodservice GPO + chain account targeting	Named GPOs (PFG Vistar, Premier, Vizient, FoodBuy, Avendra)	GPO contract + named regional chain account — adds \$2-8M/yr predictable volume at 14-18% margin
WMS + voice-pick + automation	Named platforms (Manhattan Associates, Blue Yonder, HighJump, SAP EWM, Vocollect Voice)	Vocollect Voice + WMS upgrade — drops pick error rate from 0.5% to 0.05% + drops labor cost 22%
Cold chain + temperature monitoring	Named cold-chain platforms (Sensitech, DeltaTrak, Carrier Sensitech, ORBCOMM)	Sensitech temperature monitoring + cold-chain audit — captures premium foodservice + retail accounts requiring temp documentation
Specialty / ethnic / niche food line	Named specialty distributors + market gaps	Add named ethnic line-card (Mexican, Asian, Mediterranean) — captures growing market segment + premium \$/case
Foodservice + restaurant outside sales territory	Named CRM + territory tools	Outside sales rep with named territory + \$1.4M revenue book — adds new chef + GM relationships at \$80k all-in cost
Beverage distributor: brand portfolio + craft	Named beverage brands + state franchise law	Add 4 named craft brewery brands + RTD cocktail brands — captures growing premium beverage segment at higher margin
Federal + DLA Troop Support contract	DLA Troop Support + Subsistence prime + sub	DLA Subsistence Prime Vendor + named subcontractor — captures \$14-40M annually in military foodservice work
Foodservice equipment + smallwares cross-sell	Adjacent product categories + manufacturer programs	Smallwares + cleaning / chemical cross-sell — drives 18% incremental revenue at 28% margin

Category	Research basis	Example with named entity
Private label / store-brand program	Named contract manufacturers + private label economics	Private label foodservice line — 32% margin vs 18% branded — drives margin lift across customer book
Foodservice operator e-commerce + B2B marketplace	Named B2B marketplaces (Faire, Mable, BlueCart, Notch)	BlueCart marketplace listing — captures 14-22 new restaurant accounts/mo at \$0 sales-rep cost
Sustainability + waste-reduction program	Named programs (Imperfect Foods, Too Good To Go for foodservice, named local food rescue)	Sustainability program + B Corp + food rescue partner — captures sustainability-conscious accounts (growing chain segment)
Acquisition + roll-up strategy	Named M&A; platforms + valuation multiples (6-9× EBITDA typical for foodservice distribution)	Acquire regional competitor — adds 200-600 customer accounts + line-card consolidation + warehouse synergy
Pricing + program management discipline	Named pricing platforms (PROS, Vendavo, Zilliant, Pricefx)	Pricefx pricing optimization — captures 1-3% margin lift across full book = \$480k-1.4M annual margin gain

Specificity rules (anti-hallucination)

- Every named distributor must be real (Sysco, US Foods, Performance Food Group [PFG], Gordon Food Service, US Foods, UNFI, KeHE, DPI, Reinhart, Ben E. Keith)
- Every named GPO must be real (PFG Vistar, Premier, Vizient, FoodBuy, Avendra, Foodbuy USA, Compass Foodbuy, ARAMARK)
- Every named platform must be real (BlueCart, Choco, Notch, Foodbam, eFoods Direct, Manhattan Associates, Blue Yonder, Vocollect Voice)
- Every named association must be real (IFDA, IFMA, NACS, NACDS, NCA — beverage, BWA — beverage wholesalers, NACFA)
- Every named cold-chain platform must be real (Sensitech, DeltaTrak, Carrier Sensitech, ORBCOMM, Emerson)

Cost scale calibration

Foodservice + grocery distribution spend bands operate at warehouse + fleet + per-account scale:

Band	Spend type
1 — Under \$5,000	Association membership, marketplace listing, training
2 — \$5,000 to \$50,000	CRM + ordering platform, marketing, GPO contract setup
3 — \$50,000 to \$500,000	WMS + cold chain + outside sales territory expansion
4 — \$500,000 to \$5M	New refrigerated truck fleet, warehouse expansion, M&A; small
5 — Over \$5M	Major M&A; full WMS migration, new region expansion

Persona schema

B2B_buyer (chef + GM + buyer + procurement)

- Buyer role (chef, GM, owner-operator, corporate buyer, procurement)
- Account size (single restaurant / small chain / regional chain / national)
- Decision driver (price, fill rate, product range, sales rep, credit terms)
- Order frequency + delivery cadence (3-5x weekly typical for restaurant)
- Product mix (broadline vs specialty vs ethnic)
- LTV (annual revenue × margin × retention years)

Cycle / calendar model

Foodservice + grocery distribution runs on holiday + tourism + weekly cycles:

Window	Trigger	Action
Holiday season (Nov-Dec)	Peak restaurant + grocery demand	Capacity max + delivery discipline + holiday SKU mgmt
Summer (Jun-Aug)	Tourism + outdoor dining + beverage peak	Resort + tourism account capacity
Q1 slow season (Jan-Mar)	Restaurant slow + grocery moderate	New-account onboarding + program reviews
School foodservice (Sep-May academic year)	School lunch program	School-account discipline + USDA commodity
Daily / weekly: order + delivery cycle	Multiple deliveries per account per week	Order discipline + fleet optimization
Quarterly: customer pricing + program review	Customer pricing + rebate	Pricing discipline + program audit
Trade show season	Named industry shows (NRA Show, Foodservice Expo, Natural Products Expo East/West)	Booth + customer meetings + new-line pipeline

Issue library

- Fill rate + on-time-in-full pressure — universal — frame as WMS + cold chain discipline
- Margin compression from Amazon Business + Sysco + US Foods scale — universal — frame as service moat + specialty
- Driver / warehouse labor shortage — universal — frame as automation + retention pay + named programs
- Cold chain compliance + regulatory pressure — universal — frame as monitoring + named platforms
- Bad-debt write-offs from restaurant failures — universal — frame as credit discipline
- Pricing + program complexity at scale — universal — frame as named pricing platform
- Sales rep productivity vs cost — universal — frame as territory + CRM discipline
- Customer churn / acquisition balance — universal — frame as outside + inside + marketplace mix

Report sections

Section	Contents
Distributor snapshot	Warehouse + fleet, employees, customer count, vertical mix, annual revenue, named comp set
Top 5 capability + scale moves	Drawn from 14-category playbook
Online ordering + tech roadmap	Named platforms + investment priorities
Line-card + vertical specialty plan	Named brand + ethnic + specialty opportunities
GPO + chain + named-account pipeline	Named GPO + chain prospects
Cold chain + WMS + automation roadmap	Named platforms + capital priorities
Sales territory + outside + inside coverage plan	Territory + comp + CRM discipline
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per truck + warehouse
- On-time-in-full % (target > 97%)
- Order accuracy / fill rate (target > 99%)
- Gross margin % (foodservice 16-22%, specialty 22-32%)
- Customer retention 12-mo + new-account growth
- Driver + warehouse retention
- Online ordering % of revenue
- Avg revenue per account

Engineering guidance

Sector 42 is pure B2B middle layer — invisible to consumers but huge to engineering. Build wholesale.durable_goods first (Phase 4, biggest small-business addressable, industrial + electrical + plumbing distributors are the biggest count). Foodservice + grocery distribution (42.B) is Phase 4-5 specialty (cold chain + GPO + scale dynamics differ enough to need separate profile). Both sub-clusters share substrate with construction.commercial_gc and manufacturing.industrial_b2b for B2B-relationship logic.

CHAPTER 48-49 — TRANSPORTATION & WAREHOUSING

Truckload + LTL + freight, courier + last-mile, warehousing + 3PL. Phase 4 priority — every B2B sector depends on this layer.

One-line truth

A trucking company runs on driver retention + fuel cost + DOT compliance + brokered loads, with margins of 2-8%. A 3PL warehouse sells space + labor + tech to e-commerce + manufacturing customers on multi-year contracts. A courier / last-mile operator competes with Amazon Logistics + USPS + UPS / FedEx. v3 has zero awareness of any of this.

NAICS code count: 47 **Sub-clusters:** 3

transportation.trucking_freight — Trucking + freight (truckload [TL], less-than-truckload [LTL], owner-operators, freight brokers)

Business model	B2B_transactional + B2B_relationship hybrid
NAICS examples	484110, 484121, 484122, 484220, 484230, 488510 (freight broker)
Competitor radius	Local / regional TL: 200-600 mi haul lanes. Long-haul TL: 600-2,500 mi. Regional LTL: 4-state region. National LTL: full continental US. Owner-operators: defined by lane preference + home base + book of brokers.

Data sources

Source	Returns	Cost	Fallback
FMCSA SAFER + DOT registration	Carrier license + safety scores + violations	Free public	Direct
DAT + Truckstop.com load boards	Load + lane pricing + capacity intel	Subscription	Internet Truckstop
American Trucking Associations (ATA) + state trucking association	Industry benchmarks + named peers	Member-only	Public
Named TMS + ELD platforms (McLeod, TMW, MercuryGate, Project44, KeepTruckin Motive, Samsara, Omnitracs)	Internal + benchmark data	Subscription	Direct

Source	Returns	Cost	Fallback
Named freight brokerage networks (CH Robinson, JB Hunt 360, Convoy [closed], Uber Freight, Coyote, Echo Global)	Brokerage relationships + tech	Direct	Marketplace
SAM.gov + GSA Freight Schedules	Federal freight opportunities	Free	Direct
Named factoring companies (TBS, Apex Capital, Triumph Business Capital, RTS Financial)	Cash flow + named partners	Direct	Industry
CarrierWatch + RMIS + Highway broker tools	Carrier verification + compliance	Subscription	FMCSA

Review categories tracked

- On-time pickup + delivery
- Driver professionalism + safety
- Equipment condition
- Cargo damage + claims rate
- Communication + tracking visibility
- Pricing competitiveness
- Capacity flexibility
- Documentation + paperwork accuracy

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Driver recruitment + retention pay program	Industry turnover (90%+ at large carriers) + named pay programs	\$0.04/mi pay raise + 1% safety bonus + paid time off — drops driver turnover from 110% to 38% — saves \$480k/yr at 40-truck fleet (\$12k cost per replacement)
ELD + telematics + safety platform	Named platforms (Samsara, Motive, Omnitracs, Lytx, SmartDrive)	Samsara ELD + dashcam + telematics — drops accident rate 28% + insurance premium savings + driver behavior coaching data
Insurance + EMR reduction	Named carriers (Great West, National Indemnity, Progressive Commercial, Sentry, Northland)	EMR from 1.10 to 0.78 + safety culture investment — drops workers comp + auto liability premium \$480k/yr at 40-truck fleet

Category	Research basis	Example with named entity
TMS (Transportation Management System) modernization	Named platforms (McLeod, TMW Suite, MercuryGate, Project44, AscendTMS)	McLeod LoadMaster + Project44 visibility — drives 14% revenue lift through dispatch efficiency + customer-visibility premium
Lane specialization + dedicated freight	Lane analysis + named shipper relationships	Dedicated lane contract with named shipper (e.g. food retail or industrial) — captures \$40-120k/wk recurring revenue at 12% margin (vs 3-5% spot freight)
Reefer / specialty trailer expansion	Specialty trailer economics + named industries	Add reefer trailers + cold-chain certification — captures food + pharma freight at 18% premium \$/mi vs dry van
Direct shipper relationships vs brokered loads	Direct vs brokered margin profile	Build 8 direct shipper relationships — replaces 28% brokered loads at \$0.18/mi margin lift = \$580k/yr at 40-truck fleet
Factoring + AR + cash flow management	Named factoring companies + factoring economics	Apex Capital + RTS factoring — same-day cash on invoices vs 30-45 day customer pay — drops cost-of-capital + enables growth
CDL training school partnership / in-house	Named CDL schools + apprenticeship programs	Partnership with named CDL school + in-house finishing program — guarantees 8 new qualified drivers/yr + reduces hiring cost
IRP / IFTA + multi-state operating discipline	IRP base plates + IFTA fuel tax	Multi-state IRP discipline + IFTA reporting + named software (TruckingOffice, Q7, ProTransport) — eliminates compliance penalties + audit risk
Sustainability + alternative-fuel + electric fleet	Named EV truck programs (Tesla Semi, Freightliner eCascadia, Volvo VNR Electric, Daimler eCascadia)	Pilot 2 electric trucks on local routes — captures sustainability-conscious shipper RFQs + state grant programs
Federal contract + GSA Freight Schedule	GSA Schedule + DLA Defense Transportation	GSA Freight Schedule + DLA registration — captures \$1-4M/yr federal freight at premium per-load rate
Driver app + mobile + paperless workflow	Named driver apps (Trucker Path, Mudflap, EROAD)	Driver-facing app + paperless BOL + ePOD — drives driver satisfaction + drops back-office admin time 40%
Acquisition + small-fleet roll-up	Named M&A; advisors + valuation multiples (3-6× EBITDA typical)	Acquire 8-truck competitor at 4× EBITDA — adds revenue + driver pool + customer relationships at known retention

Specificity rules (anti-hallucination)

- Every named TMS / ELD must be real (McLeod LoadMaster + PowerBroker, TMW Suite + Trimble TMS, MercuryGate, Project44, FourKites, Samsara, Motive, Omnitracs, Lytx, EROAD)
- Every named load board must be real (DAT, Truckstop.com, 123Loadboard, Direct Freight)
- Every named brokerage must be real (CH Robinson, JB Hunt 360, Uber Freight, Coyote, Echo Global Logistics, Worldwide Express, Total Quality Logistics [TQL])
- Every named insurance carrier must be real (Great West, National Indemnity, Progressive Commercial, Sentry, Northland, Hudson, Lancer)
- Every named factoring company must be real (TBS, Apex Capital, Triumph Business Capital, RTS Financial, OTR Solutions)
- Every named association must be real (ATA, OOIDA, NASTC, TIA, ATRI)

Cost scale calibration

Trucking + freight spend bands operate at fleet + tech + per-truck scale:

Band	Spend type
1 — Under \$5,000	Cert / association membership, training, software trial
2 — \$5,000 to \$50,000	ELD + dashcam fleet, factoring setup, marketing
3 — \$50,000 to \$500,000	TMS upgrade, multiple-truck add (used), specialty trailer
4 — \$500,000 to \$5M	New-truck fleet add, electric pilot, terminal acquisition
5 — Over \$5M	Fleet expansion, M&A; acquisition, multi-region expansion

Persona schema

B2B_buyer (logistics manager + shipper)

- Shipper type (manufacturer, distributor, retailer, foodservice, ag, federal)
- Decision driver (price, on-time, capacity, equipment, safety, communication)
- Acquisition channel (direct, broker, RFP, dedicated contract, marketplace)
- Lane / freight type (TL, LTL, partial, dedicated, expedited)
- Annual freight spend + carrier-base size
- LTV (annual revenue × margin × retention × cross-lane growth)

Cycle / calendar model

Trucking runs on freight-economy + harvest + holiday + diesel-price cycles:

Window	Trigger	Action
Q4 holiday peak (Oct-Dec)	Peak retail + e-commerce freight	Capacity max + premium pricing on spot freight
Produce season (May-Oct)	Ag + reefer demand	Reefer capacity + premium \$/mi
Spring (Mar-May)	Construction + manufacturing ramp	Industrial freight pickup
Diesel-price cycles	Fuel cost volatility	Fuel surcharge discipline + hedging

Window	Trigger	Action
Annual: insurance + DOT + IFTA renewal	Compliance + insurance	Renewal discipline + EMR audit
Quarterly: customer review + lane analysis	Lane profitability	Lane discipline + unprofitable freight pruning
Driver hiring cycle (continuous)	Driver shortage perpetual	Recruiting funnel discipline

Issue library

- Driver turnover + recruiting cost — universal — frame as pay + culture + tenure programs
- Diesel + fuel cost volatility — universal — frame as fuel surcharge + hedging
- Equipment / truck downtime — universal — frame as preventive maintenance + telematics
- Insurance premium increases — universal — frame as EMR reduction + safety culture
- DOT / FMCSA compliance violations — universal — frame as audit-ready + telematics
- Bad-debt + slow-pay shippers — universal — frame as factoring + credit discipline
- Spot-freight margin compression — universal — frame as direct-shipper + dedicated lanes
- Cargo damage / claims — universal — frame as driver training + proper securement

Report sections

Section	Contents
Carrier snapshot	Trucks + drivers, lanes, customer mix, annual revenue, named comp set
Top 5 capability + retention moves	Drawn from 14-category playbook
Driver recruitment + retention plan	Pay + culture + named program tactics
Tech stack + ELD + TMS roadmap	Named platforms + investment priorities
Direct-shipper + dedicated freight pipeline	Named target shippers + lane analysis
Insurance + safety + EMR program	Named carrier + benchmark targets
Equipment + sustainability roadmap	Fleet replacement + EV + alternative-fuel
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per truck
- Driver turnover % (target < 70%)

- On-time delivery % (target > 96%)
- Empty-mile % (target < 10% TL)
- Operating ratio (revenue - operating expense / revenue, target < 92)
- Cargo claim ratio
- EMR (target < 0.85)
- Direct vs brokered freight % (target growing direct)

transportation.courier_last_mile — Courier + last-mile delivery (small-package, food, medical, B2B same-day)

Business model	B2B_transactional + B2C_walk_in hybrid
NAICS examples	492110, 492210
Modern variants	Amazon DSP (Delivery Service Partner), Medical Specimen Courier, Restaurant Delivery, Same-Day B2B Logistics
Competitor radius	Hyper-local: 25-50 mi for same-day metro. Regional: 100 mi for medical / B2B. Statewide for specialty (medical specimen, legal documents). Amazon DSP / national gig: defined by metro routing zones.

Data sources

Source	Returns	Cost	Fallback
Amazon DSP program info + earnings reports	Named program data + benchmarks	Public	Direct
DoorDash + UberEats + Grubhub + Instacart driver economics	Gig + last-mile market intel	Public	Direct
Named TMS + dispatch (Onfleet, Bringg, Routific, Circuit, Optimoroute)	Internal benchmarks + tools	Subscription	Direct
CLDA (Customized Logistics & Delivery Association) + ESC	Industry benchmarks + named peers	Member-only	Public summary
FedEx + UPS + USPS + DHL public data	Named giant comp data	Public	Direct
Named medical courier networks (Quest Diagnostics, LabCorp, Mednet, ICE Mobility)	Medical courier intel	Direct	Industry
Local SAM.gov + state procurement	Federal + state same-day work	Free	Direct

Source	Returns	Cost	Fallback
Local Yelp + Google Reviews	Customer sentiment (esp. food + retail)	\$0.02	Yelp Fusion

Review categories tracked

- On-time delivery vs window
- Driver professionalism
- Damage / handling care
- Tracking + visibility
- Pricing transparency
- Customer service / issue resolution
- Coverage area + speed options
- Specialty handling (medical, food, fragile)

Opportunity playbook (12 categories)

Category	Research basis	Example with named entity
Amazon DSP (Delivery Service Partner) program	Amazon DSP economics + named partners + named contracts	Amazon DSP — start at \$30k investment + 20-40 vans + Amazon-routed packages — generates \$1-5M annual revenue at 5-10% margin (Amazon manages all customer-facing)
Route optimization software	Named platforms (Onfleet, Bringg, Routific, Circuit, Optimoroute, Routyn, Locus, Project44)	Onfleet (\$149-499/mo) route optimization — drops driver miles 22% + adds 14% more stops/day + customer-tracking experience
Medical specimen / pharmaceutical specialty	Named medical-courier networks + chain-of-custody	Quest Diagnostics + LabCorp specimen routes — captures \$14-48k/wk recurring at premium \$/mile + medically-fragile specialty positioning
Same-day B2B legal / financial / industrial document	Named B2B same-day demand + pricing	Same-day legal courier + financial document service — premium \$/run vs gig delivery + steady B2B account book
Restaurant + food delivery 3rd-party platform	DoorDash + UberEats + Grubhub + Local + Hyper-Local platforms	Restaurant delivery franchise / fleet — runs on top of DoorDash + UberEats with branded local-delivery overlay — captures premium order value + drives restaurant relationships
White-label last-mile for Shopify + DTC brands	Shopify + named DTC shipping platforms	White-label same-day for Shopify merchants — captures premium e-commerce pricing + locks merchant relationships at \$14-22 per delivery
Driver app + dispatch tech	Named driver apps + dispatch tools	In-house driver app + dispatch + customer tracking — drives driver satisfaction + customer experience differentiation vs gig competitors

Category	Research basis	Example with named entity
Independent contractor (1099) vs W-2 model	AB-5 + state-by-state IC laws + named compliance	1099 model with proper IC discipline + named compliance — drops fully-loaded labor cost 30% vs W-2 (subject to state-level constraints)
Cargo bike + electric scooter + sustainable fleet	Named e-bike + cargo-bike platforms + city zones	Cargo e-bike fleet for downtown core — captures last-mile in dense urban + congestion-charge zones + sustainability narrative
Same-day grocery + alcohol + retail partnership	Named retail same-day partnerships (Instacart, DoorDash retail, Shipt)	Direct retail partnership with named grocer — same-day fleet + dedicated driver pool — captures recurring routed work
Reverse logistics / returns processing	Named reverse logistics platforms (Happy Returns, Optoro, Loop Returns)	Returns + reverse logistics for e-commerce brands — captures growing market segment + drives carrier relationships
Subscription / contracted fleet (named accounts)	Subscription delivery economics + named partners	Subscription fleet for named B2B account (\$24k/mo for dedicated 4-driver coverage) — drives recurring revenue + driver utilization

Specificity rules (anti-hallucination)

- Every named platform must be real (Onfleet, Bringg, Routific, Circuit, Optimoroute, Locus, Project44, Bringg, Tookan)
- Every named giant must be real (Amazon DSP, FedEx Ground + Express, UPS, USPS, DHL Express + Supply Chain, OnTrac, LSO)
- Every named gig platform must be real (DoorDash, UberEats, Grubhub, Instacart, Shipt, GoPuff, Roadie, Postmates [now UberEats])
- Every named medical-courier partner must be real (Quest Diagnostics, LabCorp, BioReference, Sonic Healthcare USA, MedSpeed, Mednet)
- Every named association must be real (CLDA, ESC, NASCS, AASLD, NRA — National Restaurant)

Cost scale calibration

Courier + last-mile spend bands operate at vehicle + per-driver + tech scale:

Band	Spend type
1 — Under \$1,000	App / platform setup, training, marketing
2 — \$1,000 to \$10,000	Route optimization software, branded uniforms, magnet vehicle wraps
3 — \$10,000 to \$100,000	Vehicle add (used), driver training program, dispatch tech
4 — \$100,000 to \$1M	Multi-vehicle fleet add (new), DSP setup, sustainable fleet pilot
5 — Over \$1M	Major fleet expansion, M&A; acquisition, regional buildout

Persona schema

B2B_buyer + B2C_consumer hybrid

- Customer type (B2B account, retail merchant, individual consumer, medical/healthcare)
- Decision driver (price, speed, reliability, tracking, specialty handling)
- Volume (one-off vs subscription vs route)
- Acquisition channel (direct, marketplace, referral, partnership)
- Account integration depth (API, EDI, manual)
- LTV (avg run × frequency × retention × cross-service)

Cycle / calendar model

Courier + last-mile runs on retail + holiday + medical-routine + restaurant cycles:

Window	Trigger	Action
Q4 holiday peak (Nov-Dec)	E-commerce + retail peak	Capacity max + driver overflow + premium pricing
Spring (Mar-May)	Tax document + medical surge	Tax-document season + growing medical routes
Daily: peak hours	Lunch + dinner + commerce surges	Driver utilization + dynamic dispatch
Weekly: M-F vs weekend	Different demand patterns	Schedule flexibility + part-time supplementation
Annual: vehicle insurance + DOT + cargo van CDL exemptions	Compliance	Renewal discipline
Quarterly: customer churn + acquisition review	Account base health	Retention + new-account discipline
Annual: vehicle replacement + maintenance cycle	Fleet capex	Replacement schedule + maintenance discipline

Issue library

- Driver shortage + retention vs gig — universal — frame as pay + culture + tenure
- Vehicle wear + maintenance cost — universal — frame as PM + fleet management
- Insurance premium for last-mile — universal — frame as named carrier + safety culture
- IC vs W-2 misclassification risk — universal — frame as compliance + clear contractual structure
- Customer-platform algorithm dependency — universal — frame as direct + diversified mix
- Last-mile margin pressure from giants — universal — frame as specialty + premium positioning
- Damage + liability claims — universal — frame as training + insurance + handling protocols
- Tech / app reliability — universal — frame as platform discipline + backup

Report sections

Section	Contents
Operation snapshot	Vehicles + drivers, service area, customer mix, annual revenue, named comp set
Top 5 strategic moves	Drawn from 12-category playbook
Customer mix + diversification plan	Named B2B + B2C + retail account targets
Tech + dispatch + tracking roadmap	Named platforms + investment priorities
Driver recruiting + retention plan	Pay + culture + classification model
Specialty / vertical positioning plan	Medical / legal / B2B / DSP / e-comm pathway
Insurance + safety + compliance program	Named carrier + safety practices
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per vehicle
- On-time delivery % (target > 95%)
- Driver retention
- Stops per route
- Cost per stop / cost per delivery
- Customer retention 12-mo
- Damage / claim rate
- Direct vs platform-routed % of revenue

transportation.warehousing_3pl — Warehousing + 3PL (third-party logistics, fulfillment, public warehousing, cold storage)

Business model	B2B_relationship
NAICS examples	493110, 493120, 493130, 493190
Competitor radius	Local: 60-100 mi service area. Regional 3PL: 4-state. National 3PL: continental US. E-comm fulfillment networks: 2-day-ground reach (typically 5-12 facilities for national coverage). Specialty (cold, hazmat, FFE) regional + niche.

Data sources

Source	Returns	Cost	Fallback
IWLA (International Warehouse Logistics Association) + WERC	Industry benchmarks + named peers	Member-only	Public summary
Named WMS + 3PL platforms (Manhattan Associates, Blue Yonder, HighJump, SAP EWM, Logiwa, ShipBob WMS, Cin7, Veeqo)	Internal + tech benchmarks	Subscription	Internal
Inbound Logistics + DC Velocity + Modern Materials Handling	Industry trends + benchmarks	Subscription	Public
Named e-commerce platforms (Shopify, Amazon FBA, Walmart Fulfillment Services, Target Plus)	E-comm fulfillment intel	Public / sub	Direct
Named real estate (CoStar + JLL Industrial + CBRE + Cushman & Wakefield)	Warehouse market data	Subscription	Public summary
SAM.gov + GSA + DLA Distribution	Federal warehousing opportunities	Free	Direct
Named cold-chain certs (HACCP, SQF, BRC, AIB, FDA)	Cold-storage compliance + cert	Free / sub	Direct
CBRE + JLL named broker contacts	Building / lease + tenant intel	Direct	Industry

Review categories tracked

- Order accuracy + fill rate
- On-time fulfillment SLA
- Inventory accuracy
- Damage + claims rate
- Tech + customer portal experience
- Pricing + cost transparency
- Account management responsiveness
- Specialty handling (cold, hazmat, value-add)

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
WMS modernization + automation	Named platforms (Manhattan Associates SCALE, Blue Yonder, Logiwa, HighJump, SAP EWM, Microsoft Dynamics 365 Supply Chain, Korber, Made4Net)	Logiwa or Manhattan Associates WMS — captures 22% productivity + customer-portal + e-commerce-ready capability
Robotics + automation pilot	Named platforms (AutoStore, 6 River Systems [Shopify], Locus Robotics, Geek+, Berkshire Grey, Fetch Robotics)	Locus Robotics autonomous mobile robots + AutoStore — drops pick labor 60% on growing e-commerce volume + scales without facility expansion
E-commerce fulfillment specialty (Shopify, Amazon FBM, Walmart, Target+)	Named platforms + integration partners	Shopify-certified 3PL + ShipStation integration — captures e-commerce DTC merchants at \$4-12/order pick + pack revenue
Cold storage + temperature-controlled specialty	Named cold-chain certs + specialty pricing	Cold storage build + HACCP + SQF cert — captures food + pharma + medical premium pricing at 2x standard storage rates
Named-account 3PL contracts	Named target customer types (CPG, e-commerce, manufacturer, retailer)	Named CPG account 3-yr contract — \$2-8M/yr recurring revenue at 18% margin (vs spot warehousing)
Value-add services (kitting, assembly, returns, custom)	Value-add economics + customer demand	Kitting + custom packaging + returns — captures 38-50% margin on incremental services (vs 18% on storage + handling)
Customer portal + visibility + integration	Named integration platforms (EDI, API, Shopify, ShipStation, NetSuite)	Customer portal + real-time inventory + EDI integration with named top-10 customers — drives 22% margin lift via stickiness
Hazmat / DG (dangerous goods) capability	OSHA HAZMAT + DOT HMR + named cert	HAZMAT capability + DG-certified staff — captures premium chemicals + battery + electronics work at 28% premium \$/sqft
Federal + DLA Distribution contract	DLA Distribution + GSA + DoD warehousing programs	DLA Distribution Depot prime/sub or GSA warehousing — captures \$4-20M/yr predictable federal volume

Category	Research basis	Example with named entity
Sustainability + LEED + energy efficiency	Named programs (LEED for Warehouse, Energy Star, named utility rebates)	LED retrofit + solar + Energy Star certification — drops energy cost 28% + qualifies for sustainability-focused tenant RFPs
Multi-channel fulfillment (DTC + B2B + retail)	Multi-channel WMS + integration	Multi-channel fulfillment for hybrid customer (e.g. Shopify DTC + B2B EDI to retail) — captures higher revenue per customer + locks long-term contract
Acquisition + roll-up strategy	Named M&A; advisors + valuation multiples (8-14× EBITDA for 3PL)	Acquire complementary 3PL (specialty or geographic) — adds revenue + customer cross-sell + capability consolidation
Cross-dock + transload capability	Cross-dock economics + named customers	Cross-dock capability for retail / e-comm customers — premium \$/touch vs storage + drives faster customer revenue
Real estate + lease optimization	Named industrial RE brokers + market rates	Long-term industrial lease with named broker (CBRE, JLL, Cushman) at sub-market rate — locks cost basis + drives margin lift

Specificity rules (anti-hallucination)

- Every named WMS must be real (Manhattan Associates SCALE + Active WM, Blue Yonder, HighJump, SAP EWM, Logiwa, Microsoft Dynamics 365, Korber, Made4Net, Cin7, Veeqo, ShipBob WMS)
- Every named robotics platform must be real (AutoStore, 6 River Systems, Locus Robotics, Geek+, Berkshire Grey, Fetch Robotics, GreyOrange, OPEX)
- Every named cold-chain cert must be real (HACCP, SQF, BRC, AIB, FDA, NSF)
- Every named e-comm platform must be real (Shopify Plus + Shopify, Amazon FBA + FBM + Multi-Channel Fulfillment, Walmart Fulfillment Services, Target Plus, ShipStation, ShipBob)
- Every named association must be real (IWLA, WERC, CSCMP, NASSTRAC)
- Every named broker must be real (CBRE Industrial + Logistics, JLL Industrial, Cushman & Wakefield, Colliers Industrial, Newmark)

Cost scale calibration

Warehousing + 3PL spend bands operate at facility + tech + capital scale:

Band	Spend type
1 — Under \$10,000	Cert / association membership, training, marketing
2 — \$10,000 to \$100,000	WMS module add, cert audit, customer portal
3 — \$100,000 to \$1M	WMS upgrade, robotics pilot, cold-storage retrofit
4 — \$1M to \$10M	Major automation, facility expansion, M&A; small
5 — Over \$10M	New facility, major automation, regional expansion, M&A;

Persona schema

B2B_buyer (logistics + supply chain + e-commerce ops)

- Customer type (CPG manufacturer, e-commerce DTC, retailer, distributor, federal)
- Decision driver (cost, accuracy, speed, capability, integration, geography)
- Acquisition channel (RFP, broker, referral, marketplace, direct sales)
- Volume tier (small / medium / large / enterprise)
- Multi-year contract appetite
- LTV (annual revenue × margin × retention years × value-add upsell)

Cycle / calendar model

Warehousing + 3PL runs on retail / holiday + e-commerce-volume + customer-onboarding cycles:

Window	Trigger	Action
Q4 holiday peak (Aug-Dec)	Peak inventory + fulfillment volume	Capacity discipline + temp labor + customer SLA management
Customer onboarding cycles (rolling)	New customer integration	Onboarding process discipline + integration timing
Annual: WMS / tech roadmap	Tech investment cycle	Roadmap + named-platform planning
Annual: cert audit (HACCP, SQF, BRC, AIB)	Cert + compliance audits	Pre-audit walkthrough + management review
Quarterly: customer QBR + scorecard	Customer-managed scorecard	On-time + accuracy + cost review
Lease cycles (5-10 yr typical)	Real estate decisions	Renegotiation + capacity planning + named broker
Industry events (ProMat, MODEX, NRF Big Show)	Tech + customer + competitive intel	Event + named-prospect pipeline

Issue library

- Customer SLA + fill-rate pressure — universal — frame as WMS + automation + cycle counts
- Labor cost + warehouse worker shortage — universal — frame as automation + retention + temp ratio
- Capacity vs demand mismatch — universal — frame as multi-tenant + flexible + cross-dock
- Cert + audit failure risk — universal — frame as continuous compliance + management review
- Customer integration complexity (EDI / API) — universal — frame as named platforms + integration discipline
- Cold-chain failure risk — universal — frame as monitoring + redundancy
- Real estate cost increases — universal — frame as long-term lease + named broker
- Margin compression vs national 3PL scale — universal — frame as specialty + value-add positioning

Report sections

Section	Contents
3PL snapshot	Facilities, sqft, capacity, customer mix, annual revenue, named comp set
Top 5 capability + scale moves	Drawn from 14-category playbook
WMS + automation + tech roadmap	Named platforms + investment priorities
Customer pipeline + named-account targets	Named target customers + RFP cadence
Specialty / vertical positioning plan	Cold + hazmat + e-comm + value-add pathway
Cert + compliance roadmap	Named audit programs + cert pursuit
Real estate + lease + capacity plan	Named broker + lease + facility roadmap
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per sqft
- On-time fulfillment % (target > 98%)
- Order accuracy / fill rate (target > 99.5%)
- Inventory accuracy (target > 99.5%)
- Customer retention 12-mo (target > 90%)
- Damage / claim rate
- Cert + audit pass rate (target 100%)
- Multi-year contract % of revenue (target > 60%)

Engineering guidance

Sector 48-49 is the invisible backbone — every other sector depends on transportation + warehousing. Build transportation.trucking_freight first (Phase 4, biggest small-business count + most operationally diverse). Courier + last-mile (48-49.B) is Phase 4 (most accessible barrier-to-entry, growing market segment with Amazon DSP). Warehousing + 3PL (48-49.C) is Phase 4-5 specialty (highest LTV per customer, longest sales cycle).

CHAPTER 56 — ADMINISTRATIVE & SUPPORT / WASTE MANAGEMENT

Staffing agencies, janitorial + facility services, security + pest control, waste management. Phase 4 priority — recurring B2B service contracts at scale.

One-line truth

A staffing agency runs on candidate database + named-account books, not Yelp reviews. A commercial janitorial firm has 60-200 named property managers + facility managers as customers and competes on labor pricing + insurance + bonding. A pest-control or waste-management firm runs route density economics that v3 has zero awareness of. Four distinct B2B sub-clusters.

NAICS code count: 56 **Sub-clusters:** 4

admin.staffing_agency — Staffing agencies (light industrial, clerical, executive search, technical / IT)

Business model	B2B_relationship
NAICS examples	561311, 561312, 561320, 561330
Modern variants	IT Staffing Specialist, Healthcare Travel Nurse Agency, Executive Search, RPO Provider
Competitor radius	Local light industrial / clerical: 25-50 mi. Regional: 4-state. National + IT + executive search: national. Specialty (healthcare travel, niche): national.

Data sources

Source	Returns	Cost	Fallback
Bullhorn + Crelate + Avionté + JobDiva benchmarks	Industry ATS + CRM benchmarks	Subscription	Internal
American Staffing Association (ASA)	Industry benchmarks + named peers	Member-only	Public summary
LinkedIn Recruiter + ZipRecruiter + Indeed Resume	Candidate database + sourcing	Subscription	Direct
Named industry directories (NACE, NAPS, NTSC)	Sector-specific intel	Member-only	Public
Glassdoor + Indeed company reviews	Candidate-side reputation	Free	Direct

Source	Returns	Cost	Fallback
SAM.gov + GSA + state contracts	Federal + state staffing contracts	Free	Direct
Named MSPs (Allegis Group, ManpowerGroup, Kelly OCG, Randstad MSP)	MSP/VMS programs + named peers	Direct	Industry
Named ATS / CRM (Bullhorn, Crelate, Avionté, JobDiva, PCRecruiter)	Internal benchmarks + tools	Subscription	Direct

Review categories tracked

- Candidate quality + match
- Speed of fill / time-to-fill
- Account manager responsiveness
- Pricing transparency
- Compliance + W-2 / 1099 handling
- Candidate experience
- Retention + replacement
- Industry-specialty knowledge

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Niche industry vertical specialization	Named industry-specialty staffing firms	Healthcare travel nurse specialty + named JCAHO + state-license expertise — captures premium \$/placement + recurring weekly billables vs spot direct-hire
MSP / VMS partnership program	Named MSPs (Allegis Group, ManpowerGroup, Kelly OCG, Randstad MSP, Adecco)	MSP supplier program (e.g. Allegis, Randstad MSP) — captures named-enterprise account work + recurring billables, accepts margin compression for volume
ATS + CRM modernization	Named platforms (Bullhorn, Crelate, Avionté, JobDiva, PCRecruiter, Job Adder, Vincere)	Bullhorn + Mobile CRM + automation — drives 22% recruiter productivity + drops time-to-fill 28%
Recruiter + business development team build	Recruiter desk + BD comp economics	Hire 4 recruiters at full-desk model + 1 BD — adds \$1.2-2.4M revenue at 22% gross margin in year 2
Direct-hire / permanent placement line	Direct-hire fee structure (15-30% of salary)	Direct-hire line — average \$14-32k/placement × 80 placements/yr = \$1.5-2.5M revenue at 60% margin (vs 18% on temp)

Category	Research basis	Example with named entity
RPO (Recruitment Process Outsourcing) line	RPO economics + named programs	RPO contract with named employer — \$14k/mo retainer + per-hire fee — drives recurring revenue + locks customer relationship
Federal + state contracting (GSA Schedule, state contracts)	GSA Schedule + state contract vehicles	GSA Schedule 736 staffing services — captures \$4-12M/yr predictable federal staffing volume
Diversity / DEI specialization + cert	Named DEI cert programs (NMSDC, WBENC, NGLCC)	NMSDC MBE + WBENC WBE cert — qualifies for diversity-supplier RFPs at named Fortune 500 + federal
Workforce compliance + benefits as service	Named benefits + ACA compliance partners	In-house benefits specialist + ACA compliance — captures premium pricing + drops customer-side compliance risk
Sales / marketing + content engine	Named B2B platforms (HubSpot, LinkedIn, Apollo, Clay)	HubSpot + LinkedIn outbound + content — drives 8-14 qualified inquiries/mo at \$0 ad spend after 12 mo investment
Workers comp + General liability optimization	Named comp partners + EMR reduction	EMR from 1.10 to 0.85 + named workers comp specialist — drops fully-loaded labor cost 14% + qualifies for safety-required customer work
Niche IT + tech staffing specialty	Named IT staffing peers + specialty markets	Cybersecurity + AI / ML niche — captures premium bill rates (\$150-225/hr) + steady demand growth + low candidate competition
Acquisition + roll-up strategy	Named M&A; advisors + valuation multiples (5-9× EBITDA)	Acquire smaller staffing firm at 6× EBITDA — adds candidate pool + customer relationships + geographic / vertical extension
Healthcare credentialing + compliance specialty	Named credentialing programs (Joint Commission, NCQA, named state boards)	Healthcare credentialing capability — captures travel nurse + locum tenens premium pricing + niche moat

Specificity rules (anti-hallucination)

- Every named ATS must be real (Bullhorn, Crelate, Avionté, JobDiva, PCRecruiter, Job Adder, Vincere, Loxo, Tracker, Zoho Recruit)
- Every named MSP must be real (Allegis Group, ManpowerGroup, Kelly OCG, Randstad MSP, Adecco, Aerotek, Robert Half MSP)
- Every named association must be real (American Staffing Association ASA, NACE, NAPS, NTSC, SHRM)
- Every named diversity cert must be real (NMSDC for MBE, WBENC for WBE, NGLCC for LGBTQ, NaVOBA for VOSB, USPAACC for AAPI)
- Every named federal contract vehicle must be real (GSA MAS Schedule including 736 staffing services, named state contracts)

Cost scale calibration

Staffing agency spend bands operate at small-business + recruiter + tech scale:

Band	Spend type
1 — Under \$1,000	Association membership, training, partner application
2 — \$1,000 to \$10,000	ATS / CRM, marketing, RPO setup
3 — \$10,000 to \$100,000	Recruiter team build, niche specialty buildout, marketing
4 — \$100,000 to \$1M	Multi-recruiter expansion, second office, M&A; small
5 — Over \$1M	Major M&A;, multi-office expansion, RPO at scale

Persona schema

B2B_buyer + B2C (candidate) hybrid

- Customer type (HR director, hiring manager, procurement, MSP buyer)
- Decision driver (cost, speed, candidate quality, compliance, scale)
- Volume tier (one-off / contingent / managed contract / RPO)
- Industry vertical fit
- Acquisition channel (RFP, referral, content / inbound, MSP, direct sales)
- LTV (avg billable × hrs × retention × cross-sell)

Cycle / calendar model

Staffing runs on hiring-cycle + seasonal + economic cycles:

Window	Trigger	Action
Q1 (Jan-Mar)	New-year hiring + budget approval	Capacity max + new-account ramp
Summer slow-down (Jul-Aug)	Hiring slow + recruiter capacity	BD + outreach + new-account focus
Back-to-school + Q4 (Sep-Dec)	Holiday + seasonal staffing peak	Light industrial + retail + foodservice surge
Federal FY-end (Sep)	Federal hiring deadline	Federal + state account ramp
Annual: customer rate review + program audit	Customer rate cards	Rate discipline + program optimization
Quarterly: candidate pipeline + ATS health	Recruiter productivity + database health	Database hygiene + new-candidate sourcing
Annual: workers comp + insurance + ACA	Compliance + insurance	Renewal discipline + audit prep

Issue library

- Recruiter turnover + ramp time — universal — frame as comp + culture + ATS support

- Margin compression from MSP / VMS programs — universal — frame as direct + premium specialty
- Candidate-database staleness — universal — frame as continuous nurture + ATS discipline
- Workers comp + claims experience — universal — frame as safety culture + claims management
- Customer churn / VMS forced bidding — universal — frame as direct relationships + niche moats
- Compliance burden (ACA, EEO, classification) — universal — frame as in-house specialist + named platform
- Speed of fill vs candidate quality tradeoff — universal — frame as continuous sourcing + candidate experience
- Bill rate vs pay rate spread + visibility — universal — frame as transparent rate cards + value articulation

Report sections

Section	Contents
Agency snapshot	Recruiters, vertical mix, customer count, annual revenue, named comp set
Top 5 growth + diversification moves	Drawn from 14-category playbook
Niche specialization + vertical roadmap	Industry vertical + cert + premium positioning
Recruiter team + BD comp plan	Named structures + comp benchmarks
ATS + tech roadmap	Named platforms + investment priorities
Customer + named-account pipeline	Named MSP + direct + RPO targets
Compliance + workers comp program	Named partner + EMR target
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + GP\$ per recruiter
- Time-to-fill (target < 12 days for clerical, < 25 for skilled)
- Direct-hire % + recurring revenue mix
- Recruiter retention 12-mo
- Candidate database active count
- Customer retention 12-mo
- GP margin % (target 22-32% on temp, 60%+ direct-hire)
- EMR (target < 0.95)

admin.facility_services — Janitorial + facility services (commercial cleaning, building maintenance, landscaping commercial, multi-service)

Business model	B2B_relationship
NAICS examples	561720, 561740, 561730 (commercial landscaping), 561210 (facilities support)
Competitor radius	15-25 mi for cleaning crew dispatch economics. Regional 50-100 mi for premium / specialty (post-construction, healthcare). National multi-site contracts via national operators (ABM, ISS, Aramark, OneSource).

Data sources

Source	Returns	Cost	Fallback
BSCAI (Building Service Contractors Association International)	Industry benchmarks + named peers	Member-only	Public summary
ISSA (worldwide cleaning industry)	Industry benchmarks + cert programs	Member-only	Public
Cleaning & Maintenance Management magazine	Industry trends	Subscription	Public
Named CRM + dispatch (Aspire Software, Service Pro, ServiceTrade, Janitorial Manager, CleanGuru)	Internal benchmarks + tools	Subscription	Direct
Local property mgmt + facility mgmt directories	Named target customers	Free / member	Direct
SAM.gov + GSA + state contracts (custodial services)	Federal + state opportunities	Free	Direct
Named CIMS (Cleaning Industry Management Standard) + Green Seal	Industry cert + sustainability	Direct	Industry
Named insurance partners + bonding companies	Insurance + bonding intel	Direct	Industry

Review categories tracked

- Cleaning quality + thoroughness
- Reliability + on-time arrival
- Communication + responsiveness
- Crew professionalism + uniformed
- Pricing transparency
- Insurance + bonding documentation
- Equipment + supplies quality
- Specialty / sustainability capability

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
CIMS (Cleaning Industry Management Standard) certification	ISSA CIMS + CIMS-GB (Green Building) cert	CIMS + CIMS-GB cert — captures premium commercial RFPs requiring industry-recognized quality standards + premium pricing
OSHA 10 + 30 + bloodborne pathogen training	OSHA + named training programs	OSHA + bloodborne pathogen + Hepatitis B vaccination program — captures medical-facility + lab + research customer work
Specialty service line expansion (post-construction, biohazard, COVID disinfection)	Named adjacent service economics	Post-construction cleanup specialty — premium \$/sqft + recurring relationships with named GCs (e.g. local AGC member companies)
Healthcare / cleanroom specialty	Named healthcare cleaning standards + AAMI + ISSA training	Healthcare-specialty cleaning + AAMI training — captures hospital + ambulatory + lab work at premium pricing
Sustainability + Green Seal + named-product lines	Green Seal + LEED + named eco product lines	Green Seal certified + LEED-aligned cleaning program — captures sustainability-conscious customer RFPs + property management premium
Multi-service expansion (cleaning + landscaping + maintenance + pest)	Multi-service customer + bundle economics	Add commercial landscaping + light maintenance + pest control — captures 2.4× annual revenue per customer + drops customer-acquisition CAC
CRM + dispatch + customer portal	Named platforms (Aspire Software, Janitorial Manager, ServiceTrade, CleanGuru, Aspire — formerly LMN, ServiceWorks)	Aspire Software + customer portal — drops admin time 40% + drives customer-self-service + premium pricing
Insurance + bonding optimization	Named insurance carriers (Liberty Mutual, Travelers, Hartford) + bonding (Liberty Mutual Surety, Travelers Bond)	Insurance + bonding upgrade — qualifies for \$40M+ annual contracts requiring proof of capacity (named property management + Class A office)
Federal + GSA + state contracts	GSA MAS + GSA Schedule 03FAC + state contracts	GSA Schedule 03FAC + GSA MAS Schedule — captures \$1-4M/yr predictable federal custodial volume
Named-property-management + facility-manager pipeline	Named property mgmt cos (CBRE, Cushman & Wakefield, JLL, Colliers) + IFMA member directory	14 named property mgmt prospect campaign — adds 4-6 new commercial accounts/yr at \$48k/yr each = \$192-288k recurring
Workers comp + EMR reduction program	EMR economics + named carriers	EMR from 1.10 to 0.78 — drops workers comp 38% + qualifies for safety-required customer work

Category	Research basis	Example with named entity
Day-cleaning + sustainability + green positioning	Day-cleaning vs night-cleaning + sustainability messaging	Day-cleaning model — drops energy cost for customer (lights + HVAC during cleaning) + improves staff retention + sustainability story
Acquisition + roll-up strategy	Named M&A; advisors + valuation multiples (4-7× EBITDA)	Acquire smaller competitor with complementary customer book + drives revenue + crew + customer cross-sell
Equipment + supply + uniform program	Named equipment partners + uniform programs	Branded uniforms + commercial-grade equipment + named supply contracts (BradyPLUS, Veritiv) — drives professional appearance + drops supply cost

Specificity rules (anti-hallucination)

- Every named cert must be real (CIMS, CIMS-GB, OSHA 10/30, ISSA-CMI, Green Seal GS-37 + GS-40, LEED IEQ for Cleaning)
- Every named platform must be real (Aspire Software, Janitorial Manager, ServiceTrade, CleanGuru, ServiceWorks, ServiceMonster, Jobber)
- Every named association must be real (BSCAI, ISSA, IFMA, IREM)
- Every named insurance + bonding partner must be real (Liberty Mutual, Travelers, Hartford, Cincinnati, Federal Insurance, Liberty Mutual Surety, Travelers Bond, Chubb)
- Every named property mgmt customer must be real (CBRE, Cushman & Wakefield, JLL, Colliers, Newmark, Greystar)

Cost scale calibration

Janitorial / facility services spend bands operate at small-business + per-crew scale:

Band	Spend type
1 — Under \$500	Association membership, training, marketing template
2 — \$500 to \$5,000	CRM + dispatch software, signage, cert application
3 — \$5,000 to \$50,000	Equipment add, training, BD outreach campaign
4 — \$50,000 to \$500,000	Crew expansion, multi-service line buildout, vehicle add
5 — Over \$500,000	Acquisition, multi-state expansion, equipment fleet add

Persona schema

B2B_buyer (property + facility manager)

- Customer type (Class A office, healthcare, retail, school, industrial, federal)
- Decision driver (price, quality, reliability, insurance, specialty)
- Building size + scope (single-tenant / multi-tenant / portfolio)
- Acquisition channel (RFP, referral, walk-in, named property mgmt)
- Contract structure (monthly recurring, project-based, on-call)
- LTV (monthly contract × multi-year × cross-sell to multi-service)

Cycle / calendar model

Janitorial / facility services run on contract-renewal + seasonal + property-cycle cycles:

Window	Trigger	Action
Annual: contract renewal + RFP cycle	Most contracts annual or 2-3 yr	RFP responses + retention discipline
Construction season (Apr-Oct)	Post-construction + commercial buildout	Specialty service capacity
Q4 holiday peak (Nov-Dec)	Retail + holiday season cleaning surge	Capacity max for retail customers
Spring deep clean season (Mar-May)	Annual deep cleaning surge	Deep clean upsell discipline + scheduled service
Annual: workers comp + insurance + bonding renewal	Compliance	Renewal discipline + EMR audit
Quarterly: customer QBR + scorecard	Customer retention	Quality reviews + relationship discipline
Daily / weekly: route + crew dispatch	Operational	Dispatch discipline + quality control

Issue library

- Crew turnover + retention — universal — frame as pay + culture + named programs
- Quality consistency across crews — universal — frame as supervision + named platforms
- Margin compression from national operators — universal — frame as specialty + service moat
- Bonding + insurance capacity for big contracts — universal — frame as named partner + capacity ramp
- Customer churn at contract renewal — universal — frame as QBR + relationship management
- OSHA / safety / workers comp claims — universal — frame as safety culture + EMR target
- Multi-shift / overnight scheduling complexity — universal — frame as named platform + dispatch
- Sustainability + green-cleaning customer pressure — universal — frame as cert + named-product line

Report sections

Section	Contents
Operation snapshot	Crews, customer count, vertical mix, annual revenue, named comp set
Top 5 growth + retention moves	Drawn from 14-category playbook
Multi-service expansion plan	Named adjacent services + cross-sell
Specialty + cert positioning roadmap	Named cert + healthcare + green pathways

Section	Contents
Named-property-management pipeline	Named target customers + RFP cadence
Tech + dispatch + portal roadmap	Named platforms + investment priorities
Workers comp + insurance + bonding plan	Named partners + EMR + capacity
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per crew
- Customer retention 12-mo (target > 90%)
- Crew retention 12-mo
- Customer scorecard / quality complaints rate
- Multi-service customer % (target > 35%)
- GP margin % (target 18-28%)
- EMR (target < 1.0)
- Insurance + bonding capacity (\$)

admin.security_pest — Security services + pest control (residential + commercial pest, termite, security guard, alarm monitoring)

Business model	B2B_relationship + B2C_walk_in hybrid
NAICS examples	561612, 561621, 561622, 561710 (pest), 561730 (lawn / pest crossover)
Competitor radius	Pest control: 25-50 mi route density radius. Termite + WDO: 50-100 mi specialty. Security guard: 30-60 mi metro. Alarm monitoring: national (Central Station) + local (response). National operators (Orkin, Terminix, ADT, Allied Universal) compete in every market.

Data sources

Source	Returns	Cost	Fallback
NPMA (National Pest Management Association)	Pest industry benchmarks + named peers	Member-only	Public summary
ASIS International	Security industry benchmarks + cert	Member-only	Public
ESA + named alarm-monitoring centers	Alarm industry intel	Member / direct	Industry

Source	Returns	Cost	Fallback
Named CRM + dispatch (PestPac, Field Routes, Briostack, ServSuite, Service Autopilot)	Internal benchmarks + tools	Subscription	Direct
Local Yelp + Google Reviews (residential)	Customer sentiment	\$0.02	Yelp Fusion
Pest Control Technology magazine	Industry trends	Subscription	Public
SDM magazine (Security Distributing & Marketing)	Security trends	Subscription	Public
Named state licensing boards (pest control + security)	License + violations	Free public	Direct

Review categories tracked

- Effectiveness of treatment
- Tech professionalism + uniformed
- On-time arrival + windows
- Pricing transparency + recurring schedule
- Communication + reporting
- Pet + child safety claims
- Specialty pest knowledge (termite, bed bug, rodent)
- Response time on emergency / repeat

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Quarterly recurring service plan	Recurring revenue economics + named structures	Quarterly pest plan @ \$89/qtr × 1,200 customers = \$427k/yr recurring at 38% margin (vs spot-treatment 18%)
Specialty: termite / WDO + warranty	Named termite warranty + bonding programs	Termite warranty + \$250k repair guarantee — premium \$/treatment + drives customer retention + real-estate transaction work
Bed bug specialty + heat treatment	Bed bug treatment economics + named equipment	Bed bug heat treatment specialty + named equipment (ThermaPure, GreenTech) — premium \$1,400-3,200/treatment + low competition
Commercial + multifamily contract pipeline	Named property mgmt + commercial accounts	40 commercial accounts × \$480/mo = \$19.2k/mo recurring at 42% margin (vs residential 32%)

Category	Research basis	Example with named entity
CRM + route optimization	Named platforms (PestPac, FieldRoutes, Briostack, ServSuite, Service Autopilot, Jobber, ServiceTitan)	PestPac or FieldRoutes — captures 22% productivity + customer self-service + automated billing
Sustainability + IPM (Integrated Pest Management) positioning	Named IPM cert + sustainability messaging	Green IPM + Green Pro + EcoSmart certifications — captures sustainability-conscious customer segment + premium pricing
Mobile / online sales + customer portal	Named platforms + customer experience	Online sales + mobile-first signup — captures 28% of new customers after-hours + drops sales-rep cost
Acquisition + roll-up strategy (high-margin)	Named M&A; advisors + valuation multiples (8-14× EBITDA — pest control is high-multiple sector)	Acquire smaller pest control firm at 10× EBITDA — drives revenue + customer cross-sell + route density
Security guard service add-on	Combined pest + security + facility services	Add security guard service for commercial customers — captures cross-sell at premium pricing + locks customer
Alarm + monitoring service add-on	Named alarm partners (Bay Alarm, Tyco SimpliSafe, ADT, Brinks)	Alarm + monitoring partnership — adds \$32-48/mo recurring per customer + drives customer retention
Reputation + review velocity program	Named platforms (Birdeye, Podium, NiceJob)	NiceJob automated review request — Google rating from 4.0 to 4.7 + 4× review velocity
Lead-platform investment (Google LSA, Angi, HomeAdvisor)	Named lead platforms + cost per booked job	Google LSA + Angi Pro — \$25-65 cost per booked customer + Google Guaranteed badge premium
State + local commercial licensing maturation	Named cert programs (state + ASIS for security, ACE + ESA for pest)	ACE (Associate Certified Entomologist) + state cert + safety training — drives premium positioning + commercial RFP qualification
Tech-driven pest monitoring (smart traps, IoT)	Named platforms (Anticimex SMART, Bell Sensing, Spotta)	Anticimex SMART smart pest monitoring — premium B2B commercial offering + recurring data fees

Specificity rules (anti-hallucination)

- Every named platform must be real (PestPac, FieldRoutes [now ServiceTitan], Briostack, ServSuite, Service Autopilot, Jobber, ServiceTitan, Marathon Data PestPro)
- Every named cert must be real (NPMA, ACE — Associate Certified Entomologist, GreenPro, EcoSmart, ASIS for security, named state pest licenses, OSHA, IPM)
- Every named giant must be real (Orkin, Terminix, Rollins, Rentokil, Massey Services, Truly Nolen, Western Exterminator)

- Every named alarm + security must be real (ADT, Vivint, Brinks, Tyco SimpliSafe, Allied Universal, G4S, Securitas, Bay Alarm)
- Every named association must be real (NPMA, ASIS International, Electronic Security Association ESA, Pest Control Technology PCT)

Cost scale calibration

Pest + security spend bands operate at small-business + per-route scale:

Band	Spend type
1 — Under \$500	Cert application, association membership, training
2 — \$500 to \$5,000	CRM + dispatch software, lead-platform setup, signage
3 — \$5,000 to \$50,000	Truck + equipment, route expansion, marketing campaign
4 — \$50,000 to \$500,000	Multi-truck add, second branch, IoT specialty buildout
5 — Over \$500,000	Acquisition, multi-state expansion, fleet expansion

Persona schema

B2C_consumer (residential) + B2B_buyer (commercial)

- Customer type (homeowner, renter, property mgmt, commercial, restaurant)
- Decision driver (price, effectiveness, sustainability, brand, schedule)
- Service type (one-off, quarterly, monthly, on-demand emergency)
- Acquisition channel (Google, neighbor referral, walk-by truck, HomeAdvisor, named property mgmt)
- Specialty pest issue (termite, bed bug, rodent, ant, roach)
- LTV (monthly recurring × retention years × upsell)

Cycle / calendar model

Pest control runs on weather + pest-emergence + season cycles:

Window	Trigger	Action
Spring (Apr-Jun)	Ant + termite + flea / tick surge	Capacity max + commercial + residential surge
Summer (Jun-Aug)	Mosquito + wasp + bee + general pest	Specialty mosquito service + emergency
Fall (Sep-Nov)	Rodent + spider + winter prep	Rodent + winterization service push
Winter (Dec-Mar)	Slow season + indoor pest	Bed bug + indoor specialty + planning
Quarterly: recurring schedule	Quarterly customer service	Route discipline + quality consistency
Annual: license + cert + insurance	Compliance	Renewal discipline + audit prep
Termite season (warmer months)	Termite emergence + WDO	Termite + WDO inspection + treatment

Issue library

- Pest re-infestation / treatment failure — universal — frame as IPM + warranty + retreatment policy
- Pesticide safety + family / pet exposure — universal — frame as Green / IPM + named-product safety
- Tech turnover impact on customer relationships — universal — frame as named-tech retention + customer-tech matching
- Margin compression from national operators (Terminix, Orkin) — universal — frame as service moat + relationship
- Slow-pay residential customers — universal — frame as autopay + credit-card discipline
- Customer churn at low-engagement quarterly — universal — frame as comm cadence + value reporting
- Specialty pest + emergency capacity — universal — frame as named specialty positioning + on-call rotation
- License + state regulation complexity — universal — frame as compliance discipline + named platform

Report sections

Section	Contents
Operation snapshot	Trucks + techs, customer count, route density, annual revenue, named comp set
Top 5 growth + retention moves	Drawn from 14-category playbook
Specialty + recurring revenue plan	Named specialty + recurring tier opportunities
Commercial + multifamily pipeline	Named property mgmt + commercial prospects
Tech + CRM + dispatch roadmap	Named platforms + investment priorities
Lead + reputation channel mix	Named platforms + tactic-by-platform ROI
Customer personas (4 + lost customer)	Residential / commercial / specialty / emergency
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per truck / route
- Recurring revenue % (target > 60%)
- Customer retention 12-mo (target > 80%)
- Tech retention 12-mo
- Customer acquisition cost by channel
- Avg revenue per customer
- Treatment effectiveness rate (callback < 4%)
- Avg Google + HomeAdvisor rating

admin.waste_management — Waste management (commercial waste hauling, residential collection, recycling, demolition debris, hazardous waste)

Business model	B2B_relationship + B2C municipal hybrid
NAICS examples	562111, 562112, 562119, 562211, 562213, 562910 (remediation)
Competitor radius	Local: 30-60 mi service area for route economics. Regional 4-state for specialty (hazardous, demolition, C&D;). National giants (WM, Republic, Waste Connections) compete in every market via acquired locals.

Data sources

Source	Returns	Cost	Fallback
NWRA (National Waste & Recycling Association)	Industry benchmarks + named peers	Member-only	Public summary
SWANA (Solid Waste Association of N. America)	Industry trends + benchmarks	Member-only	Public
Waste Today + Waste 360 + Recycling Today	Industry intel + benchmarks	Subscription	Public
Named giants benchmark (Waste Management WM, Republic Services, Waste Connections, Casella)	Public competitive intel	Public	Direct
EPA + state DEQ	Compliance + permit data	Free public	Direct
Named CRM + routing (Soft-Pak, Roadnet by Roper, Encompass Waste, AMCS, Tower)	Internal benchmarks + tools	Subscription	Direct
Local municipal contracts + RFPs	Named muni opportunities	Free	Direct
SAM.gov + GSA waste services	Federal opportunities	Free	Direct

Review categories tracked

- Reliability + on-schedule pickup
- Cleanliness of containers + truck
- Pricing transparency vs surcharges
- Customer service + responsiveness
- Recycling + sustainability options
- Specialty waste handling (hazardous, e-waste, medical, demolition)
- Container condition + replacement

- Billing accuracy

Opportunity playbook (13 categories)

Category	Research basis	Example with named entity
Recycling + sustainability service line	Named recycling programs + commodity markets	Single-stream recycling + named MRF (Material Recovery Facility) partnership — captures sustainability-conscious customer + premium \$/lift + commodity revenue
Commercial + named-account pipeline	Named property mgmt + commercial accounts	Commercial sales rep + 40 named-account campaign — adds 4-6 commercial accounts/yr at \$480-2,400/mo each
Specialty waste lines (medical, hazardous, e-waste, C&D; demolition debris)	Named specialty waste licenses + permit	C&D; + roll-off specialty — captures growing construction-driven demand + premium \$/load vs standard MSW
Roll-off + temporary container service	Named contractor accounts + project-based	Roll-off container fleet (40 yard) + named contractor relationships — drives \$14-32k/wk project-based revenue at premium pricing
Routing + GPS + telematics platform	Named platforms (Soft-Pak, Roadnet by Roper, AMCS, Encompass Waste, Routeware)	Soft-Pak + Routeware GPS routing — drops empty miles 18% + adds 14% capacity + customer-tracking premium
Municipal RFP + contract pipeline	Named muni contract structures + RFP cycles	Win named municipal contract — \$1.4-4.8M/yr predictable revenue at 14-22% margin (vs commercial subject to churn)
Landfill + transfer station ownership	Vertical integration economics + named permit	Acquire transfer station / landfill — drops disposal cost 22% on own routes + adds external tipping fee revenue
EV / CNG / RNG truck fleet	Named EV + CNG + RNG truck programs (Mack LR Electric, Volvo VNR Electric, named CNG conversion)	CNG fleet + RNG (renewable natural gas) — drops fuel cost 28% + sustainability narrative + qualifies for state grants
Diversion + zero-waste consulting line	Named zero-waste cert programs + customer demand	Zero-waste consulting + diversion service for named ESG-focused customers — premium pricing + retention moat
Customer portal + autopay + e-billing	Named customer experience tools	Customer portal + autopay + e-billing — drops billing dispute 40% + drives recurring revenue + autopay enrollment

Category	Research basis	Example with named entity
Insurance + safety + EMR reduction	Named carriers + EMR economics	EMR from 1.10 to 0.78 + named safety culture program — drops insurance premium \$480k+/yr at 30-truck fleet
Acquisition + tuck-in strategy	Named M&A; advisors + valuation multiples (8-14× EBITDA — waste is high-multiple sector)	Acquire smaller hauler at 10× EBITDA — drives route density + customer cross-sell + landfill volume
Federal + state remediation contracts	EPA Superfund + state cleanup programs	Specialty environmental remediation capability — captures EPA + state-funded cleanup work at premium pricing

Specificity rules (anti-hallucination)

- Every named platform must be real (Soft-Pak, Roadnet by Roper, AMCS, Encompass Waste, Routeware, Tower Software, AssetWorks)
- Every named giant must be real (Waste Management [WM], Republic Services, Waste Connections, Casella Waste Systems, GFL Environmental, Waste Pro USA, Stericycle [medical waste])
- Every named association must be real (NWRA, SWANA, IFAT, ISWA)
- Every named regulator must be real (EPA, state DEQ + DNR, RCRA, OSHA, DOT)
- Every named alternative-fuel program must be real (named CNG fleet partners, RNG, BNEF transit programs)

Cost scale calibration

Waste management spend bands operate at fleet + landfill + capital scale:

Band	Spend type
1 — Under \$5,000	Association membership, training, marketing
2 — \$5,000 to \$50,000	CRM + routing software, customer portal, container
3 — \$50,000 to \$500,000	Truck add (used), CNG conversion pilot, route expansion
4 — \$500,000 to \$5M	Truck fleet add (new), transfer station retrofit, M&A; small
5 — Over \$5M	Major M&A;,, landfill acquisition, regional expansion, full fleet replacement

Persona schema

B2B_buyer (commercial + property mgmt) + municipal

- Customer type (residential muni, multifamily, commercial, industrial, contractor, healthcare)
- Decision driver (price, reliability, sustainability, specialty capability, scope)
- Volume (single bin / multifamily / industrial roll-off)
- Acquisition channel (RFP, referral, drive-by, named property mgmt)
- Contract structure (monthly recurring, project-based, municipal multi-year)
- LTV (monthly recurring × multi-year × cross-sell to specialty)

Cycle / calendar model

Waste management runs on construction season + holiday + municipal-RFP cycles:

Window	Trigger	Action
Construction season (Apr-Oct)	Roll-off + C&D; peak	Roll-off fleet capacity + contractor relationships
Q4 holiday peak (Nov-Dec)	Retail + commercial waste surge	Capacity max + holiday route discipline
Spring cleanup season (Mar-May)	Residential + commercial cleanup	Spring service campaigns + roll-off rental
Annual: customer rate review + RFP cycle	Pricing + program cycle	Rate discipline + RFP responses
Annual: insurance + DOT + EMR audit	Compliance	Renewal + safety culture review
Daily / weekly: route + dispatch	Operational	Route discipline + RFID + GPS
Quarterly: customer QBR + scorecard	Customer retention	QBR + relationship management

Issue library

- Driver shortage + retention — universal — frame as pay + culture + named programs
- Margin compression from national giants — universal — frame as service moat + named-relationship
- Container condition + replacement complaints — universal — frame as proactive cleaning + replacement schedule
- Pricing surprise / fuel surcharges — universal — frame as transparent surcharge + customer comm
- Recycling commodity volatility — universal — frame as diversified outlet + named processor partners
- Insurance premium increases — universal — frame as EMR reduction + named carrier
- Landfill capacity / tipping cost — universal — frame as vertical integration + named MRF partner
- Municipal contract loss / churn — universal — frame as RFP discipline + multi-year terms

Report sections

Section	Contents
Operation snapshot	Trucks + drivers, customer count, route density, annual revenue, named comp set
Top 5 growth + scale moves	Drawn from 13-category playbook
Specialty waste line plan	Named specialty + commercial + roll-off pathway
Routing + tech + customer portal roadmap	Named platforms + investment priorities

Section	Contents
Municipal + commercial named-account pipeline	Named muni + commercial prospects
Sustainability + alternative-fuel + recycling plan	Named programs + capital priorities
Insurance + safety + EMR program	Named carriers + EMR target
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per truck
- Customer retention 12-mo (target > 90%)
- Driver retention 12-mo
- Route density (stops per route)
- Recurring revenue % (target > 80%)
- Insurance EMR (target < 0.85)
- Recycling diversion %
- Equipment utilization %

Engineering guidance

Sector 56 has 4 sub-clusters with distinct B2B-relationship + recurring-revenue patterns. Build admin.facility_services first (Phase 4, biggest small-business count + most visible for lead-gen). Pest + security (56.C) is Phase 4 (high-multiple-EBITDA segment + most M&A; activity). Staffing (56.A) is Phase 5 specialty (highest-LTV but most regulated + complex). Waste (56.D) is Phase 4-5 specialty (highest capital intensity but highest-multiple acquisition value).